

Communicating acidity:

**Investigating the relationship between language use, chemical composition, and
temporal sensory perception of Riesling and Chenin Blanc**

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Table of Contents

ABSTRACT	1
1. INTRODUCTION	3
2. BACKGROUND AND CONTEXT	5
2.1 DESCRIBING WINE ACIDITY	5
2.2 ACID COMPOSITION IN GRAPES	6
2.3 ACIDITY PERCEPTION	7
2.4 TEMPORAL METHODS IN SENSORY SCIENCE	9
3. METHODOLOGY	10
3.1 WINE SOURCING	11
3.1.1 <i>Methods</i>	11
3.1.2 <i>Outcomes</i>	11
3.2 CHEMICAL ANALYSIS	13
3.2.1 <i>Methods</i>	13
3.2.2 <i>Outcomes</i>	13
3.3 SENSORY ANALYSIS	15
3.3.1 <i>Methods</i>	15
3.3.2 <i>TI data processing</i>	15
3.3.3 <i>Outcomes</i>	16
3.4 ONLINE QUESTIONNAIRE	18
3.4.1 <i>Methods</i>	18
3.4.2 <i>Descriptor data processing</i>	18
3.4.3 <i>Outcomes</i>	19
3.5 EXPERT TASTING	20
3.5.1 <i>Methods</i>	20
3.5.2 <i>Outcomes</i>	22
3.6 STATISTICAL ANALYSIS	24
3.6.1 <i>RQ1 data analysis</i>	24
3.6.2 <i>RQ2 data analysis</i>	25
3.6.3 <i>RQ3 data analysis</i>	26
4. RESULTS	27
4.1 RQ1	27
4.1.1 <i>Insights from the online questionnaire</i>	27
4.1.2 <i>Insights from the expert tasting study</i>	29
4.2 RQ2	32
4.2.1 <i>Result of TI analysis</i>	32
4.2.2 <i>Relating TI with Chemical Analysis</i>	33
4.3 RQ3	34
4.3.1 <i>Predicting descriptor-matching scores</i>	34
4.3.2 <i>Visualising language-chemistry-sensory relationships</i>	38
5. DISCUSSION	40
5.1 RQ1	40
5.1.1 <i>Theoretical Implications</i>	40
5.1.2 <i>Practical Implications</i>	41
5.2 RQ2	43
5.2.1 <i>Theoretical Implications</i>	43
5.2.2 <i>Practical Implications</i>	44
5.3 RQ3	45
5.3.1 <i>Theoretical Implications</i>	45
5.3.2 <i>Practical Implications</i>	47
5.4 LIMITATIONS AND FUTURE WORK	48

6. CONCLUSIONS.....	51
REFERENCES	53
APPENDICES	59
APPENDIX A. ACCEPTED VERSION OF RPP.....	59

Abstract

Acidity is a key structural element in wine, yet there is little scientific understanding of how people use descriptors to communicate acidity. This study investigates the relationship between language use, chemical composition, and temporal sensory perception in Riesling and Chenin Blanc, two grape varieties known for high acidity. The research aims to address three questions: (1) how wine experts describe the quality of acidity in these wines and whether variety-based differences exist, (2) how the temporal profile of acidity is affected by grape variety and chemical composition, and (3) how the interplay of chemical, perceptual, and psychological factors influences the use of acidity descriptors.

Six wines were analysed chemically, including four from established regions (Alsace and Clare Valley Riesling; Vouvray and Swartland Chenin Blanc) and one of each variety from Marlborough. Time Intensity (TI) analysis, measuring the evolution of acidity in the mouth, revealed that Riesling wines reached peak acidity more quickly and with greater intensity than Chenin Blanc wines. An online questionnaire assessing experts' mental representations of the four established styles identified eight key descriptors used in a subsequent in-person blind-tasting: *racy*, *crisp*, *fresh*, *steely*, *bracing*, *sharp*, *zesty*, *broad*. While experts clearly differentiated descriptor use by wine style in the questionnaire, during blind tasting, all terms except *broad* were more strongly matched with Riesling. Multivariate regression then revealed that descriptor use was significantly predicted by both acid and non-acid components (e.g., ethanol, CO₂), temporal dynamics, and olfactory access (e.g., wearing a nose clip led to decreased use of *racy* and *bracing*).

Taken together, the use of acidity descriptors is shaped by wine composition, temporal sensory perception, and cognitive expectations. Experts drew on clear mental constructs between wine styles and descriptor associations in a conceptual setting, but these distinctions largely collapsed during blind tasting, where descriptor use became more constrained and perceptually driven, being shaped by acid and non-acid components as well as aroma availability. These findings challenge static notions of acidity and offer practical guidance for wine education and communication: for example, by incorporating acid-specific reference tastings (e.g., malic vs. tartaric), structured sip-hold-spit exercises to capture temporal dynamics, and training that encourages students to articulate perceptual nuance rather than rely on rote descriptors. Acknowledging and embracing the multisensory complexity of acidity can thus help wine professionals describe wine with greater precision, clarity, and consistency.

1. Introduction

Acidity is a key structural component of wine, but there exists limited scientific support for how people use descriptors to communicate acidity. Although people may intuitively use specific acid descriptors, such as *broad* for Chenin Blanc and *steely* for Riesling, it is unknown what lies behind such multisensory associations. The aim of this research is to link language use with sensory perception and chemical composition (Fig. 1) to investigate how wine experts describe acidity and how these descriptors relate to perception and chemical composition. From a theoretical perspective, it contributes to the study of wine cognition by uncovering implicit drivers behind descriptor choice. Practically, this research offers new ways to train students in perceiving and articulating acidity, with the goal of improving how it is described and communicated.

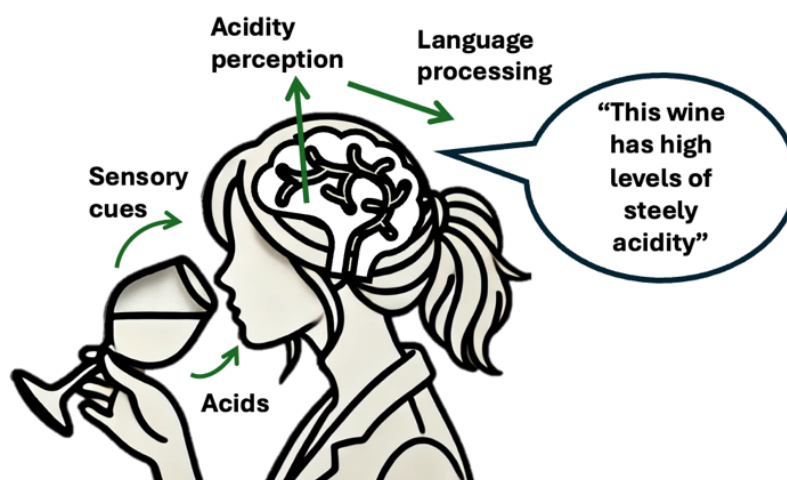


Figure 1. An illustration of various processes underlying the pathway from perception to description (candidate's own work).

Research Questions:

RQ1: How do wine experts use adjectives to describe the quality of acidity in Riesling and Chenin Blanc wines, and are there consistent variety-based differences?

RQ2: How is the wine's temporal acidity profile affected by grape variety and chemical composition?

RQ3: How does the relative relationship between chemical, perceptual, and psychological factors influence an expert's use of adjectives to describe acidity?

2. Background and context

2.1 Describing wine acidity

Tasters are taught to assess acidity in different ways. Students from the Wine & Spirit Education Trust (WSET) learn to evaluate the level of acidity from low to high (WSET, 2023), while MW students are also encouraged to state the quality of acidity (IMW, 2021; IMW, n.d.). The quality of acidity is conveyed via *descriptors*, and these adjectives often blend sensory judgement with multisensory metaphors, such as those referring to vision (e.g., *bright* acidity), space (e.g., *broad* acidity), shape (e.g., *sharp* acidity), or touch (e.g., *hard* acidity). For example, Tim Wildman MW teaches that Riesling acidity tends to be crisp, racy, and steely, while Chenin Blanc tends to have broad, firm, and tart acidity (winetutor.tv). Nick Jackson MW, in his book *Beyond Flavour* (Jackson, 2022), takes acidity description one step further by proposing the concept of acid shape, which describes the temporal evolution of acidity level in a single sip, from attack to the finish. According to Jackson, each grape variety has a signature shape that tends to remain constant regardless of origin.

Despite these existing educational references, no scientific research has offered any support for, or explanations of, any consistent use of acidity descriptors for different grape varieties. There is, however, a wide body of literature on the use of multisensory metaphors or crossmodal correspondences that have been consistently applied to describe wine (e.g., Crichton-Fock et al., 2023; Heatherly et al., 2019; Spence et al., 2013; Wang et al., 2024). For example, Wang et al. (2024) found that, when 50 novice participants evaluated the same wine using pairs of multisensory metaphors, they were able to agree at a statistically significant level, on scales of round-angular, fast-slow, and loud-quiet. This suggests that, far from random, there

may be certain aspects of the wine's chemical composition or taste experience that are driving the consistent use of these metaphorical descriptors.

2.2 Acid composition in grapes

Acidity is a crucial component of grape composition, with tartaric and malic acids being the primary acids present (Jackson, 2020). Tartaric acid is the most abundant and remains stable throughout grape ripening, while malic acid decreases as grapes ripen, due to metabolic respiration processes within the berries (Coombe, 1992). This balance between tartaric and malic acids significantly influences the acidity level in the resulting wine (Kliewer, 1967; Coombe, 1992). Other organic acids, such as citric acid, occur in smaller amounts but still contribute to the overall acid profile of the grape.

Differences in acid composition can be attributed to several factors, including climate, viticultural practices, and grape variety. Cooler climates tend to produce grapes with higher acidity, partly because slower ripening helps preserve malic acid. In warmer regions, however, rapid sugar accumulation during ripening has a greater impact on acidity loss than malic acid degradation alone. As sugars accumulate and water content increases, the resulting dilution effect lowers the concentration of organic acids in the grape, contributing significantly to the drop in total acidity (Jones & Davis, 2000). Vineyard management practices like irrigation and canopy control can also play a role in influencing acid levels (Lopes et al., 2011; Reynolds et al., 1996). For instance, irrigation can affect the rate of berry development, and shading from the vine canopy can reduce sugar accumulation and limit the degradation of malic acid, allowing for higher acid retention in the grape (Crippen & Morrison, 1986).

In addition to climate and viticultural practices, hereditary factors can affect grape acid content (Kliewer et al., 1967). For example, while Riesling and Chenin are

both high-acid varieties, Riesling is inherently higher in tartaric acid, whereas Chenin Blanc is proportionally higher in malic acid (Kliewer et al., 1967). Therefore, it is possible that any differences in variety-specific acidity descriptors could feasibly come from differences in acid composition.

Finally, it should be acknowledged that winemaking methods can have a significant impact on the overall level as well as the acid composition of the final wine (Bird & Quillé, 2022). For example, acidification can be performed via biological (e.g., specialty yeast strains), chemical (e.g., calcium sulphate in Jerez), or physical means (e.g., blending with higher acid must). Similarly, deacidification can also be carried out via biological (e.g., using lactic acid bacteria in malolactic conversion), chemical (e.g., calcium carbonate), and physical means (e.g., ion exchange column). In general, these techniques are employed to achieve desired flavour profiles or to compensate for variations in grape acidity caused by environmental or varietal factors (Jackson, 2020).

2.3 Acidity perception

Acidity perception is the sum of contributions from the wine and the taster. Wine contains a matrix of different organic acids, which is determined by grape variety as well as harvest ripeness and winemaking methods (Jackson, 2020). From the perspective of the taster, acidity perception can be ascribed to a combination of physiological (e.g. sour taste receptors, salivation) and psychological factors (e.g., taste-taste and aroma-taste interactions) (Keast & Breslin, 2003; Auvray & Spence, 2008).

Physiologically, the detection of acidity is mediated by sour taste receptors, primarily located on the tongue's taste buds, which respond to the presence of hydrogen ions in acids (Chandrashekar et al., 2006; Smith & Margolskee, 2001).

Acidity also stimulates salivation, as the presence of acids prompts the body to produce saliva in order to help neutralise it (Helm et al., 1982). This salivary response is one way to assess acidity, as higher salivation is often associated with a higher level of acidity in the wine (Schuster, 2000). Moreover, acidity perception can also be enhanced by the presence of dissolved CO₂, as CO₂ leads to the activation of sour-taste sensing receptors, and carbonic acid forms when CO₂ dissolves in liquid, producing a sharp, tingling sensation (Chandrashekar et al., 2009).

Psychologically, acidity can be influenced by both taste-taste and aroma-taste interactions. Taste-taste interactions occur when the presence of one taste suppresses or enhances the perceived intensity of another taste; medium to high concentrations of sweetness suppress acidity ("masking"), whereas low concentrations of saltiness and bitterness enhance acidity (Keast & Breslin, 2003). Aroma-taste interactions also influence acidity perception, as retronasal smell and taste combine to create the perception of flavour (Auvray & Spence, 2008). Lifelong associative learning links specific aromas and tastes, so those aromas commonly associated with sour foods, such as lemon, are perceived as 'smelling sour'. As a result, adding such sour-smelling aromas to food can enhance its perceived sourness (Wang et al., 2024).

Recent research has shown that different descriptors are associated with different acid solutions (Honoré-Chedozeau et al., 2024). In a blind tasting with 110 French oenologists, equal concentration solutions of tartaric, malic, citric, and lactic acid were tasted, and experts could select from a list of twelve descriptors to describe each solution. Correspondence Analysis revealed that tartaric acid was most associated with *astringency* and *metallic*; lactic acid with *roundness*, *bold*, and *lactic*; malic acid with *apple* and *freshness*; and citric acid with *lemon*. These results suggest

that some of the differences in the use of acidity descriptors for different wines could reflect the acid composition of grapes.

2.4 Temporal methods in Sensory Science

In Sensory Science, the level of acidity in food and beverages can be measured either by its absolute intensity or by its temporal evolution in the mouth (Lawless & Heymann, 2010). There is a large family of temporal sensory methods, including repeated ratings of sensory intensity at discrete time intervals (“Progressive Profiling”), continuous intensity tracking (“Time Intensity”), or simultaneous evaluation of several sensory attributes (e.g., “Temporal Choose All That Apply”) (Lawless & Heymann, 2010). Where there is a single sensory attribute of interest, Time Intensity (TI) is an appropriate method because it offers a complete picture of how the attribute’s intensity develops over time as the product is processed in the mouth (Peyvieux & Dijksterhuis, 2001). TI has been widely used in sensory analysis of wine to evaluate both in-mouth and aftertaste sensations (e.g., Goodstein et al., 2014, Wang et al., 2017), as it captures subtle changes in taste and flavour intensity over time: differences that are often missed by static assessment methods.

Put together, this project aims to investigate factors behind wine experts’ use of acidity descriptors in Riesling and Chenin Blanc, two of the most common high-acid white grape varieties, using both instrumental/analytical and psychological/sensory methods.

3. Methodology

This Research Paper is structured via interconnected studies which work together to comprehensively address the three research questions (Fig. 2). There are two analytical tasks followed by three experimental studies involving human subjects. The project was approved by the University of Copenhagen Faculty of Science Research Ethics Committee (case number 504-0484/24-5000).

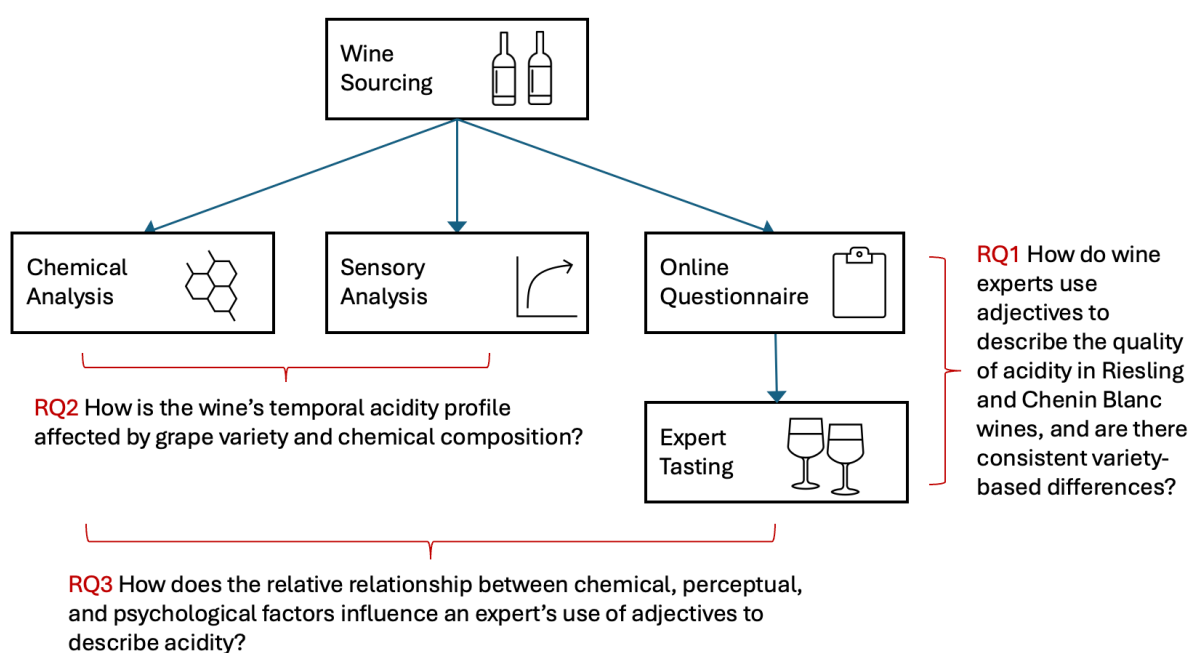


Figure 2 Research Paper structural overview

This section outlines each study's methodology and immediate outcomes, including descriptive statistics summarising the data. Inferential analyses addressing the Research Questions are presented in the Results section, while discussions on the results are presented in the Discussion section. Figure 3 gives an overview comparing the participants, samples, methods, and format of data collected across the experimental studies.

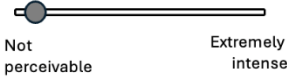
	Sensory Analysis	Online Questionnaire	Expert Tasting
Participants	N = 13 Trained panellists	N = 67 MW and MW students, globally	N = 44 DipWSET and higher, Copenhagen and London
Samples	6 wine samples Served blind in triplicate (18 samples evaluated)	4 hypothetical wine styles - Riesling from Alsace and Clare Valley - Chenin Blanc from Vouvray and Swartland	6 wine samples Served blind, once with nose clips and once without (12 samples evaluated)
Methods	Indicate the intensity of acidity you taste in the sample for each moment in time 	List up to 3 descriptors you commonly use for this wine style 	On a scale from 0 (no match at all) to 100 (matches extremely well), how well does each word describe this wine's acidity? Racy  ... Zesty 
Data collected per question	0-100 rating for each second over 75 seconds	Up to 3 words per question	0-100 ratings for 8 descriptors

Figure 3 Methods overview of the three experimental studies.

3.1 Wine sourcing

3.1.1 Methods

A total of six wine samples were selected (Table 1). High acid varieties were chosen to focus on acid *quality*, thus controlling for acid *level*. Riesling and Chenin Blanc (CB) were selected because they are major international varieties (Anderson & Nelgen, 2020), have different acid compositions (Kliwer et al., 1967) and distinctive “acid shapes” (vertical for Riesling, crescendo for CB, Jackson, 2022). While all styles are made with these varieties, the present paper focused on dry still wines (preferably residual sugar < 5 g/L) to minimise acidity interference effects from sweetness or carbonation (Heymann & Ebeler, 2017). Similarly, to encourage comparability, wines without malolactic conversion or maturation in new oak were sought after.

3.1.2 Outcomes

For each variety, two representative origins of dry wines (i.e., frequent use in the last 15 years’ MW exams) were selected. This was Clare Valley and Alsace for

Riesling, and Vouvray and Western Cape for Chenin Blanc. This allowed for coverage of stylistically distinct expressions commonly encountered by wine experts. The selection of specific wines and vintages was based on simultaneous availability in both Denmark and UK¹, and producers who had been used in previous MW exams were prioritised. To help isolate the effect of grape variety from regional or stylistic variation, a third region of Marlborough, NZ was selected, where a single producer (Astrolabe) produces both Riesling and Chenin Blanc using similar vinification techniques². Although Marlborough is not traditionally associated with these varieties, this controlled pairing offered a unique opportunity to reduce confounding variables and compare acidity expression across varieties.

Table 1. The six wine samples selected for this project, used for chemical analysis, sensory analysis, and expert tasting. Due to limited availability, not all wines are from the same vintage.

Variety	Region	Cuvee	Vintage
Chenin Blanc	Marlborough	Astrolabe Wrekin Chenin Blanc	2023
Chenin Blanc	Swartland	Mullineux Kloof Street Old Vine Chenin Blanc	2022
Chenin Blanc	Vouvray	Domain Huet Le Haut Lieu Sec	2022
Riesling	Alsace	Hugel Classic Riesling	2022
Riesling	Clare Valley	Jim Barry Lodge Hill Riesling	2022
Riesling	Marlborough	Astrolabe The Farm Dry Riesling	2023

¹ Availability in both Denmark and UK was a criterion because while sensory and chemical analysis was conducted in Denmark, expert tasting was conducted in both Denmark and UK.

² There are few producers producing dry wines from both varieties under the same climatic and winemaking conditions, so there were plausible few candidates even before considering accessibility in both DK and UK markets.

3.2 Chemical Analysis

3.2.1 Methods

The six wine samples were analysed via WineScan™ 3 (FOSS, Hillerød, Denmark), an advanced infrared spectroscopy platform for evaluating analytical parameters (Table 1). Key metrics included acidity (pH, total acidity as tartaric acid equivalent (TA), volatile acidity, citric acid, malic acid, tartaric acid, lactic acid), sugar (fructose, glucose, reducing sugar), glycerol, ethanol, CO₂, and SO₂ (free and total). To check for measurement consistency, one additional replicate sample (Vouvray) was included. Each bottle was freshly opened, and approximately 10 mL was analysed.

3.2.2 Outcomes

The results of the chemical analysis can be seen in Table 2. Overall, Alsace and Clare Valley Rieslings were highest in TA and lowest in pH. In contrast, the Swartland and Vouvray Chenin Blancs were lowest in TA. The two wine samples from Marlborough contained the highest levels of malic acid, and were unexpectedly also highest in residual sugar³, pointing to a winemaking choice to balance against malic acid. As expected⁴, the four wines bottled under screwcap also had the highest levels of CO₂.

Table 2. The most prominent analytical parameters as measured by WineScan™ 3, including those A) directly related to acidity and B) not directly related to acidity. VA = volatile acidity. Citric, malic, tartaric, lactic refer to acids. TA = total acidity expressed as tartaric acid. Glyc. = Glycerol. Red. Sugar = reducing sugar. Ethanol expressed in % vol.

³ Chemical analysis revealed that three of the wines in fact contained residual sugar slightly higher than 5 g/L. By the time of in-house chemical analysis, it was too late to re-source wines especially those from Marlborough which were custom sent to Denmark. This limitation is discussed in more detail in section 5.4.

⁴ Wines sealed with screwcaps are expected to have higher levels of CO₂ because the lower oxygen transmission rate of screwcaps makes it more difficult for CO₂ to escape (Vidal et al., 2017).

A)

Sample:	VA g/L	Citric g/L	Malic g/L	Tartaric g/L	Lactic g/L	TA g/L	pH
Marlborough Chenin Blanc 2023	0.23	0.46	4.09	1.19	0.12	8.01	3.25
Swartland Chenin Blanc 2022	0.29	0.35	2.16	2.11	0.43	6.25	3.54
Vouvray Chenin Blanc 2022	0.39	0.26	0.7	2.5	0.83	6.54	3.17
Vouvray Chenin Blanc 2022 (Replicate)	0.39	0.24	0.68	2.49	0.85	6.55	3.18
Alsace Riesling 2022	0.35	0.35	1.77	3.42	0.01	8.39	2.92
Clare Valley Riesling 2022	0.26	0.41	2.76	2.82	0.08	8.66	2.9
Marlborough Riesling 2022	0.33	0.35	3.43	1.89	0.06	8.18	3.04

B)

Sample:	Fructose g/L	Glucose g/L	Glyc. g/L	Red. Sugar g/L	Ethanol % vol.	CO₂ mg/L	Free SO₂ mg/L	Total SO₂ mg/L
Marlborough Chenin Blanc 2023	6.34	0.53	6.7	6.2	12.76	1103	13.8	69.6
Swartland Chenin Blanc 2022	0.23	0.48	6.37	1.22	12.72	1088	21.6	121.1
Vouvray Chenin Blanc 2022	5.87	0.42	7.49	6.11	13.15	755	5.7	80.6
Vouvray Chenin Blanc 2022 (Replicate)	5.94	0.27	7.49	6.08	13.13	748	5.9	80.7
Alsace Riesling 2022	3.7	0.04	6.8	3.92	12.83	977	15.6	80.5
Clare Valley Riesling 2022	2.45	0.73	5.87	3.49	12.03	1214	16	118.1
Marlborough Riesling 2022	9.56	1.28	6.1	10.4	10.7	1002	5.4	43.3

3.3 Sensory Analysis

3.3.1 *Methods*

A sensory panel of 13 people was recruited to perform the study from the staff and students at the University of Copenhagen (number of participants in line with a typical trained sensory panel, Heymann & Ebeler, 2017). To capture the fine-grained shifts in acidity intensity, participants were trained on the Time Intensity (TI) method over three 1-hour sessions. The TI method involved participants evaluating the intensity of perceived acidity continuously over time (Goodstein et al., 2014), first for 45 seconds while holding a mouthful of wine, followed by 30 seconds after spitting to assess acidity on the finish. These time periods were determined based on both prior literature (Peyvieux & Dijksterhuis, 2001; Sokolowsky & Fischer, 2012) and pilot testing with two experienced sensory panel leaders. Each 25 mL wine sample was served at room temperature (20°C) in a small plastic cup labelled with a three-digit random code. Participants were instructed to taste the entire sample to maintain procedural consistency.

During the first session, participants first calibrated acidity ratings by training on two tartaric acid and water solutions at 1.25 g/L (low concentration) and 3.0 g/L (high concentration). Then, they practised TI methodology on iPads (Compusense software, Guelph, Canada) by rating perceived visual brightness intensity of a fireworks video. In the second session, participants evaluated the six wine samples in the sensory booth. During the third session, each participant first had a warm-up sample of a different wine, and then evaluated the six wine samples in triplicate in random order.

3.3.2 *TI data processing*

In line with standard TI analysis, the curve itself is not directly assessed; instead, inferential statistics is to be conducted via extracted metrics: Area under the

curve (AUC), maximum intensity ($I_{\max}$), and time to $I_{\max}$ ($T_{\max}$) (Peyvieux & Dijksterhuis, 2001; Goodstein et al., 2014). AUC measures total perceived intensity over time, $I_{\max}$ represents the peak acidity level during each trial, and $T_{\max}$ indicates the time to reach $I_{\max}$. Combined, these metrics describe the shape of the acidity curve experienced over time (see Figure 4 for an illustration of these metrics in the context of an idealised TI curve).

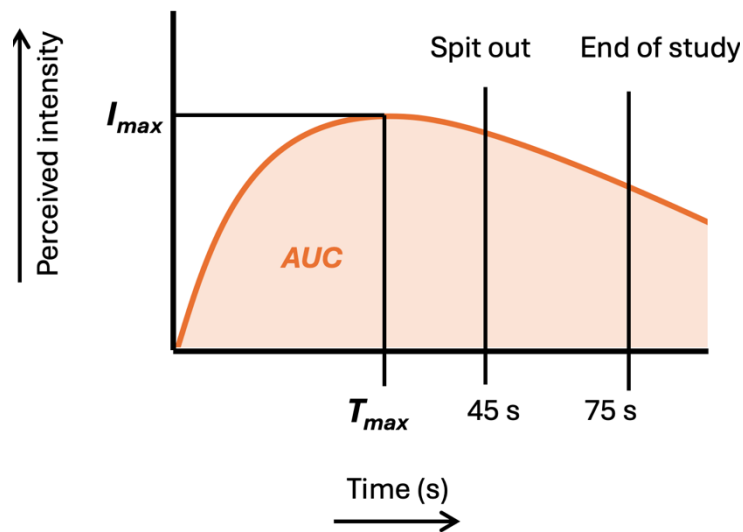


Figure 4. A TI curve with metrics of interest shown in bold italics, adapted from Goodstein et al., 2014. This is an idealised illustration not meant to reflect actual data.

3.3.3 Outcomes

Thirteen participants (8 female, average age 29 ± 4.9 years) formed the sensory panel. Each panellist evaluated the six wine samples in triplicate, totalling 18 samples per person. For visualisation purposes, Figure 5 illustrates the averaged and smoothed TI curves. More importantly, the extracted metrics for the six wines are shown in Table 3. Results of planned analysis using these outcomes are shown in Section 4.2.1, where inferential statistics are conducted using AUC, $I_{\max}$ and $T_{\max}$.

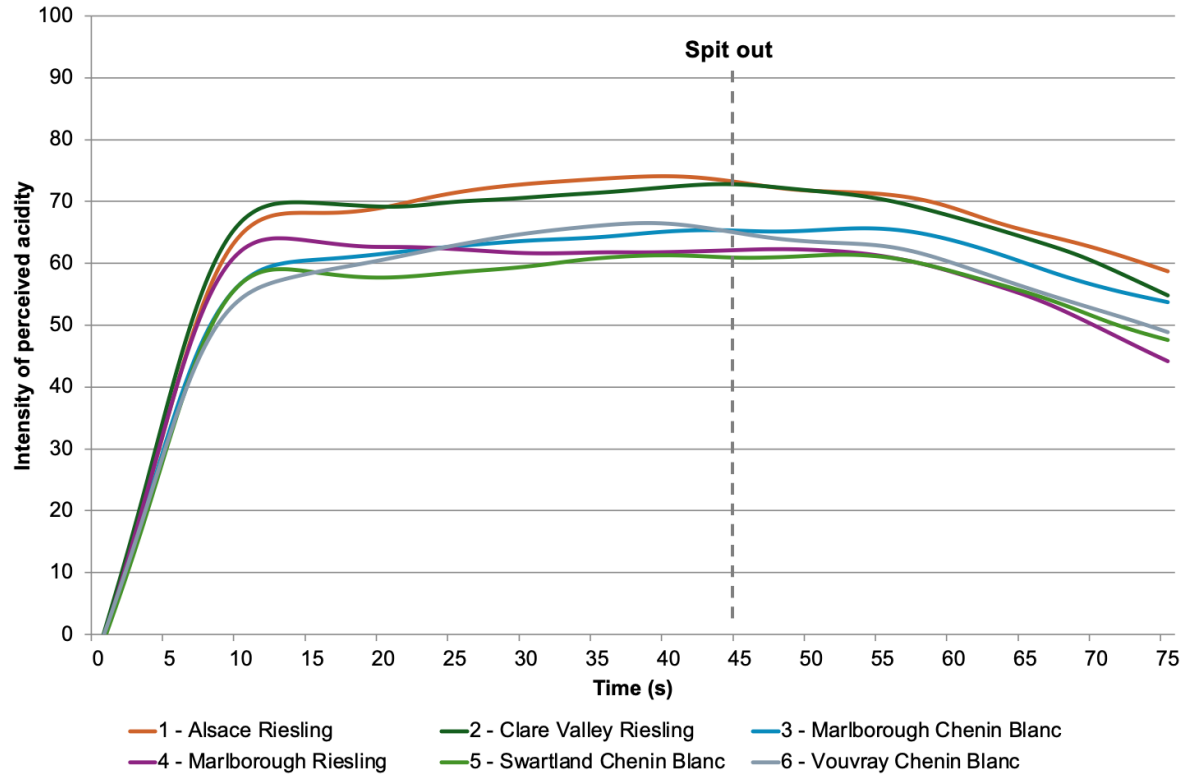


Figure 5 Smoothed TI curves across 13 panellists and 3 replicates of each wine. This represents averaged values for visual reference.

Table 3. Average extracted TI metrics by wine variety and region. Standard deviation (SD) denoted in parentheses ().

Wine	Region	AUC (SD)	lmax (SD)	Tmax (SD)
Chenin Blanc	<i>Overall</i>	4260.62 (19.57)	72.17 (15.84)	38.84 (19.57)
Chenin Blanc	Swartland	4112.41 (21.03)	70.46 (16.42)	38.59 (21.03)
Chenin Blanc	Vouvray	4281.36 (17.71)	72.13 (18.46)	39.46 (17.71)
Chenin Blanc	Marlborough	4388.10 (20.30)	73.92 (12.23)	38.46 (20.30)
Riesling	<i>Overall</i>	4664.71 (18.81)	78.08 (16.95)	28.78 (18.81)
Riesling	Alsace	4898.90 (17.13)	79.56 (15.24)	30.90 (17.13)
Riesling	Clare Valley	4850.15 (21.28)	82.00 (15.32)	29.85 (21.28)
Riesling	Marlborough	4245.08 (17.83)	72.67 (19.04)	25.59 (17.83)

3.4 Online Questionnaire

3.4.1 Methods

To identify common acidity descriptors for Riesling and Chenin Blanc to be used in the expert tasting study, an online questionnaire was launched. Participants were asked to list typical acidity descriptors for dry Riesling and Chenin Blanc from four representative regions identified earlier (Table 1). Notably, wines from Marlborough were not included in the questionnaire, as they represent a relatively unknown region for the target grape varieties. After all, the aim of the questionnaire was to produce a final list of eight adjectives based on the most common descriptors for each variety, to be used in the expert tasting.

MWs and MW students were recruited from the candidate's network and MW and MW student newsletters. To minimise cohort bias, participants were recruited from at least three continents, and with MWs graduating both before and after 2020⁵. Recruitment was guided by a power calculation (alpha = .05, power = .8, and large effect Cohen's $W = 0.5$ with 5 degrees of freedom) suggesting 52 participants.

3.4.2 Descriptor data processing

The most often used descriptors for each grape variety and each wine type were tabulated after all descriptors were converted to their stemmed forms. Stemming is a common text preprocessing step to reduce various morphological variants of a word to a single root form (e.g., zesty and zestiness would both become "zesti").

⁵ This distinction was made as it was in 2020 that Nick Jackson MW published the first version of his book *Beyond Flavour*, in which he discussed acidity shapes in detail. The assumption here is that MWs who graduated before 2020 would not be as influenced by the book, compared to those who graduated after.

3.4.3 Outcomes

A total of 67 participants (38 female, average age 42 ± 10 years) were recruited for the online questionnaire. There were 17 MWs (6 graduated after 2020) and 50 MW students (Stage 1/2/3: 9/25/16), coming from multiple continents, with the most from the Americas and Europe (Africa:1, Americas: 24, Asia: 3, Europe: 37, Oceania: 2).

Overall, a total of 148 unique descriptors in their stemmed forms were provided across all four wine styles. Figure 6 shows the most used descriptors. The final list of eight descriptors to be used for the expert tasting study was chosen based on the highest rate of total citations (**racy, crisp, fresh, steely, bracing, sharp**), followed by two descriptors often selected either for Riesling (**zesty**) and for Chenin Blanc (**broad** was chosen rather than the literal *mouthwatering*, to keep the focus of the study on metaphorical descriptors).

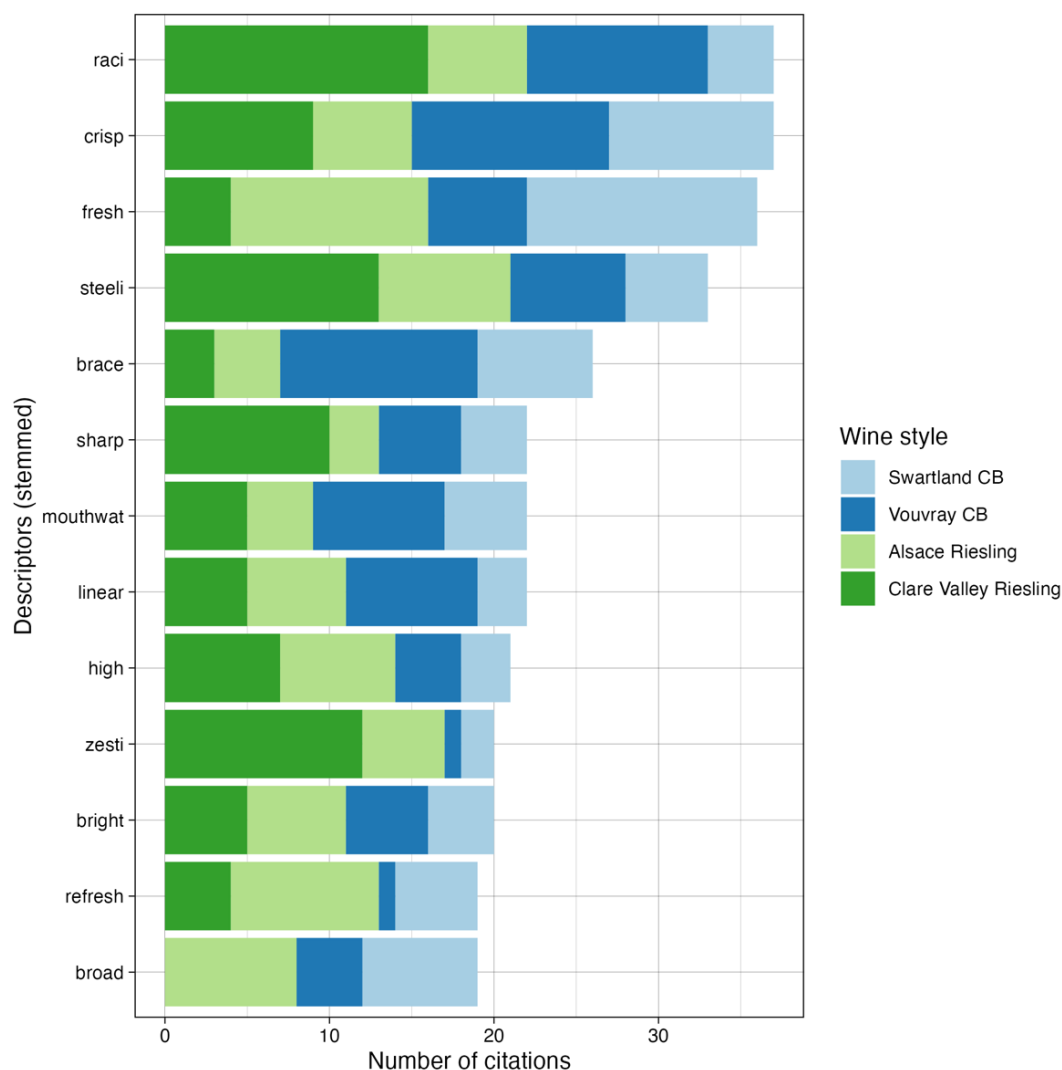


Figure 6. Overview of the most often cited descriptors (in their stemmed forms) across all four wine styles, ordered by decreasing total number of citations. Descriptors are shown in their stemmed forms.

3.5 Expert Tasting

3.5.1 Methods

To assess the extent to which experts assign common acidity descriptors to the selection of Riesling and Chenin Blanc wines, an in-person tasting experiment was conducted with participants holding the WSET Diploma certification as a minimum. The aim was to recruit 50 people (power calculation with small to medium effect size

partial $\eta^2 = 0.04$ yields 42 participants). The study took place at the University of Copenhagen and the WSET London school.

Upon arrival at the testing location, participants provided informed consent. They were then presented with eight acidity descriptors to be used during the tasting and asked to rate their familiarity with each descriptor using a visual analogue scale ranging from 0 (not familiar at all) to 100 (extremely familiar). The main experiment consisted of a blind tasting conducted in two blocks. In the first block, participants tasted wines while wearing nose clips to eliminate the influence of retronasal olfaction. In the second block, wines were tasted without nose clips to examine the influence of aroma on acidity descriptions. Following the tasting session, participants provided demographic information, including age, gender, English fluency, and their status as MW or MW students.

This study used the same six wine samples as for the chemical and sensory analyses (Table 1). The six wines were served blind in random order. For each wine, participants first tasted it, then rated how well the wine's acidity matched each of the eight descriptors identified in the online study (section 3.4). The rating was performed on a visual analogue scale from 0 (not match at all) to 100 (matches extremely well). This scale was chosen for its intuitive alignment with percentage-based reasoning and was displayed with visual guidance at 10-point intervals. Each wine was served at room temperature (20°C) in 25 mL portions in ISO glasses and labelled with a random 3-digit ID to avoid biasing participants (Fig. 7).



Figure 7 Expert tasting study layout

3.5.2 Outcomes

A total of 44 participants (22 female) completed the expert tasting study, which met the requirements of the power calculation. The average age was 42 ± 10 years. Besides holding WSET Diploma certifications, 19 participants were MW students, and 5 were MWs. Regarding English proficiency, 25 were native speakers, 8 at C2 level, 10 at C1 level, and 1 at B2 level, based on the Common European Framework of Reference for Languages (CEFR 3.3). C1 and C2 levels refer to proficient users, and B2 to high-level independent language users.

Figure 8 shows participants' self-reported familiarity level with each descriptor. Overall, participants were quite familiar with all descriptors, with average ratings above 70/100 for all descriptors. Visual inspection shows that *racy*, *broad*, and *bracing* were the least familiar, while *crisp*, *fresh*, *zesty*, and *sharp* were the most familiar.

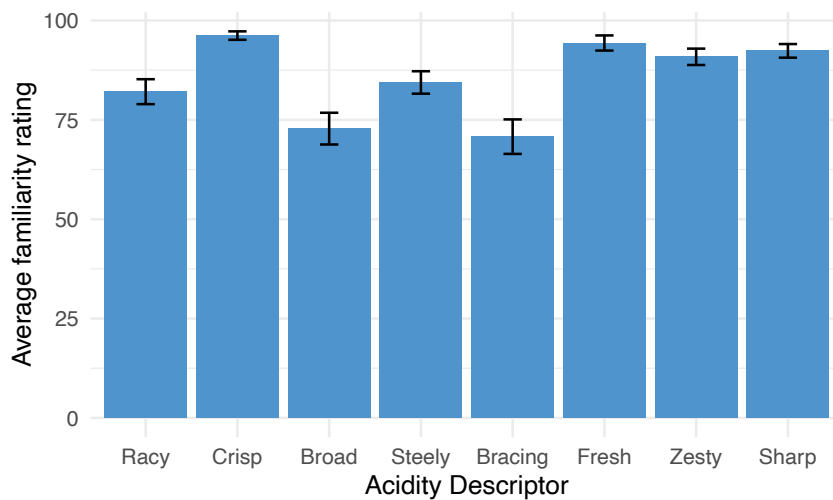


Figure 8 Average ratings of participants' self-reported familiarity with each descriptor. Error bars indicate standard error of the mean.

Figure 9 illustrates the average descriptor match ratings for each wine between conditions. Several descriptors were generally judged to be a better match across all wines with a minimum 50/100 rating, equivalent to a moderately good match (e.g., *crisp*, *fresh*, *zesty*). Other descriptors such as *racy*, *broad*, and *sharp* had varied match ratings depending on the wine. In-depth inferential statistics is presented in section 4.1.2.

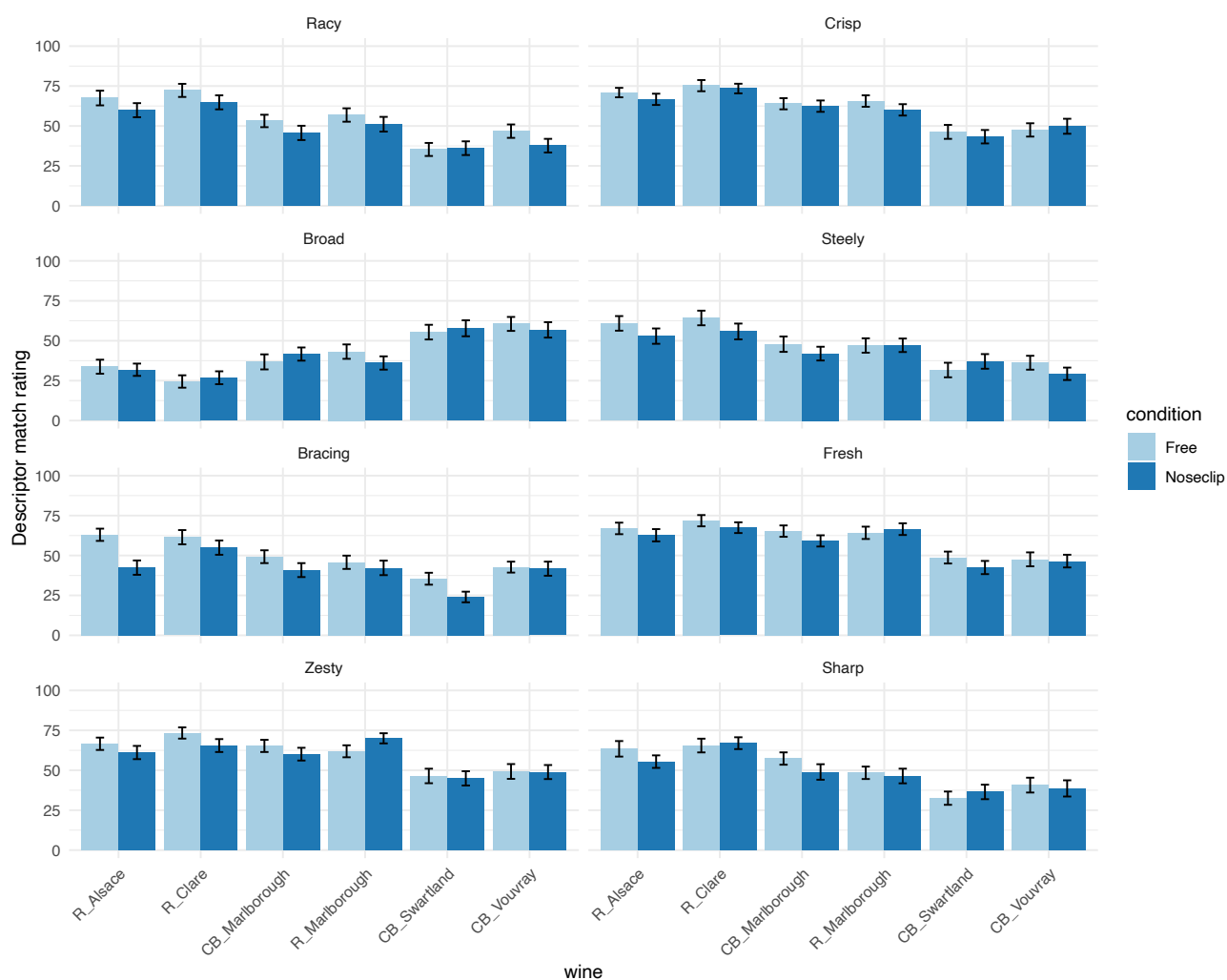


Figure 9 Average ratings of how well each descriptor matched each wine, in both tasting conditions with and without wearing a nose clip. Error bars indicate standard error of the mean.

3.6 Statistical Analysis

This section outlines the data analysis methodologies used to address the three Research Questions. Since the analyses often involve data from multiple individual studies, they are presented in a separate section here. All analyses were conducted by the candidate using the statistical software R (R Core Team 2018).

3.6.1 RQ1 data analysis

Two different methods were used to analyse how wine experts use adjectives to describe the quality of acidity in Riesling and Chenin Blanc. The first method used

results from the online questionnaire, and the second used participants' acidity descriptor-matching scores in the expert tasting.

Analysis 1: Using results from the online study, a chi-square goodness of fit test was conducted to assess whether specific descriptors were spontaneously named at greater than chance levels for specific wines. Results were then visualised with Correspondence Analysis, conducted on a contingency table with counts of acidity descriptors as rows and wine type as columns, to better understand the multivariate relationship between descriptors and wines.

Analysis 2: Using results from the expert tasting, an Analysis of Variance (ANOVA) was conducted with adjective and grape variety as independent variables and adjective rating as the dependent variable. This analysis would uncover, for each grape variety, which descriptors have significantly higher ratings. Additionally, a principal component analysis (PCA) was conducted to visualise multivariate relationships between wine samples and adjective ratings.

3.6.2 RQ2 data analysis

Two different sets of analyses were carried out to fully answer the question of how a wine's temporal acidity profile is influenced by grape variety and chemical composition. Both analyses were conducted at the aggregate level of grape variety and not on each specific wine, to address the research question regarding variety in general.

Analysis 1: Using data from the sensory analysis study, a series of ANOVAs were conducted with grape variety as the independent variable and various TI metrics (see Section 3.3.2) as dependent variables, to assess overall differences between grape varieties in terms of temporal acidity perception.

Analysis 2: Combining data from the chemical analysis and sensory analysis, correlations were conducted to elucidate the relationship between chemical composition and TI metrics.

3.6.3 RQ3 data analysis

To understand how the relationship between chemical, perceptual, and psychological factors influences an expert's use of adjectives to describe acidity, data from multiple studies were combined. For each descriptor, multivariate regression analysis was used to predict the descriptor-matching score from the expert study, based on each wine's chemical analysis, sensory analysis, and tasting condition (nose clip vs. full retronasal olfaction). A PCA then visualised the multivariate relationships between descriptors and chemical/sensory data.

4. Results

This section contains the results of inferential data analysis that directly answer the Research Questions. In line with scientific reporting conventions, results are presented here, while critical analyses of the results are reported in the Discussion section.

4.1 RQ1

How do wine experts use adjectives to describe the quality of acidity in Riesling and Chenin Blanc wines, and are there consistent variety-based differences?

4.1.1 Insights from the online questionnaire

The most frequently mentioned descriptors for each type of wine are shown below (Table 4). Notably, Chi-square tests of independence revealed that *bracing* was used significantly more for Chenin Blanc than Riesling ($\chi^2 = 5.45$, $p = .019$), and *zesty* for Riesling ($\chi^2 = 11.91$, $p < .005$)

Table 4. Top three most frequently listed acidity descriptors by wine variety and region. Frequency of citations included in parentheses (). ** indicate descriptors which are named at greater than chance levels ($p < .05$) in one variety compared to the other variety, according to chi-square tests of independence.

Wine	Region	Descriptor 1	Descriptor 2	Descriptor 3
Chenin Blanc	Overall	Crisp (22)	Fresh (20)	Bracing (19) **
Chenin Blanc	Swartland	Fresh (14)	Crisp (10)	Bracing (7)
Chenin Blanc	Vouvray	Bracing (12)	Crisp (12)	Racy (11)
Riesling	Overall	Racy (22)	Steely (21)	Zesty (17) **
Riesling	Alsace	Fresh (12)	Refreshing (9)	Broad (8)
Riesling	Clare Valley	Racy (16)	Steely (13)	Zesty (12)

Correspondence Analysis (CA) was conducted to visualise the relationships between wine styles and named descriptors (Fig. 10). Principal component 1 explains

53.18% of the variance and likely represents a range of acidity intensity, with lower perceived intensity descriptors (*round, soft*) on the positive end and higher intensity (*austere, searing*) on the negative end. Principal component 2 explains 27.84% of the variance, with more abstract descriptors (*electric, bracing*) on the positive end and fruitier descriptors (*tangy, citrusy*) on the negative end. The four wine types span the space evenly, indicating that wine experts, at least those in the MW programme, tend to use different acidity descriptors for these wines (see Section 5.4 for a discussion of participant biases). Shown in Figure 10, Clare Valley Riesling is often described by *austere, searing, zesty* acidity, while Alsace Riesling is more often described via *refreshing* and *mineral* acidity. Vouvray Chenin Blanc is most often associated with *crescendo*⁶, *electric*, and *bracing* acidity, whereas Swartland Chenin Blanc is often associated with *broad, round, juicy* acidity. Interestingly, words like *piercing, bright*, and *firm* seem to be used equally likely for all four types of wines.

⁶ This is in line with the assignment of Chenin Blanc to crescendo acidity shape according to Nick Jackson MW (Jackson, 2022).

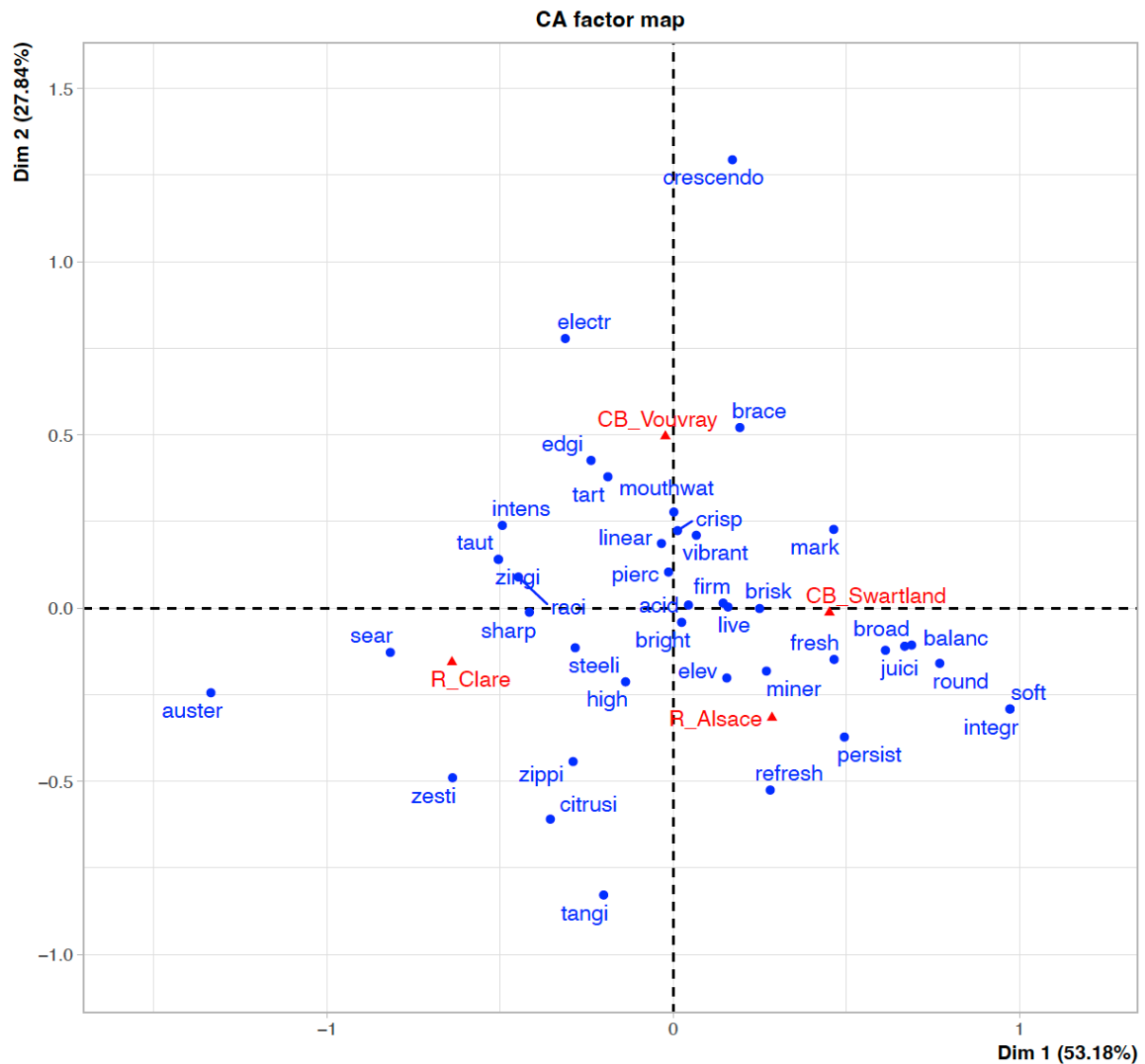


Figure 10. Correspondence Analysis of counts of descriptors in their stemmed forms across each of the four wine types in the online questionnaire. R = Riesling, CB = Chenin Blanc. Proximity of descriptors (blue) and wines (red) indicates the likelihood of descriptors being used to describe that wine. Descriptors which are located close to each other tend to be used together. The distance between wines indicates how distinct or similar the wines are based on associated acidity descriptors.

4.1.2 Insights from the expert tasting study

In the expert study, participants tasted actual wines rather than imagining typical styles. Each participant rated how well each of the eight acidity descriptors matched each wine. ANOVA results revealed a significant interaction effect between

grape variety and acidity descriptors ($F(7,4165) = 30.19, p < .005$), indicating that the two grape varieties were consistently associated with different descriptors. Post-hoc testing with Tukey pairwise multiple comparisons revealed that, with the six wines tasted grouped by variety, Riesling wines were rated with a higher match rating than Chenin Blanc wines for the descriptors *racy*, *crisp*, *steely*, *bracing*, *fresh*, *zesty*, and *sharp* ($p < .005$ for all comparisons). Only the descriptor *broad* was rated as a better matching descriptor for Chenin Blanc instead of Riesling (Fig. 11).

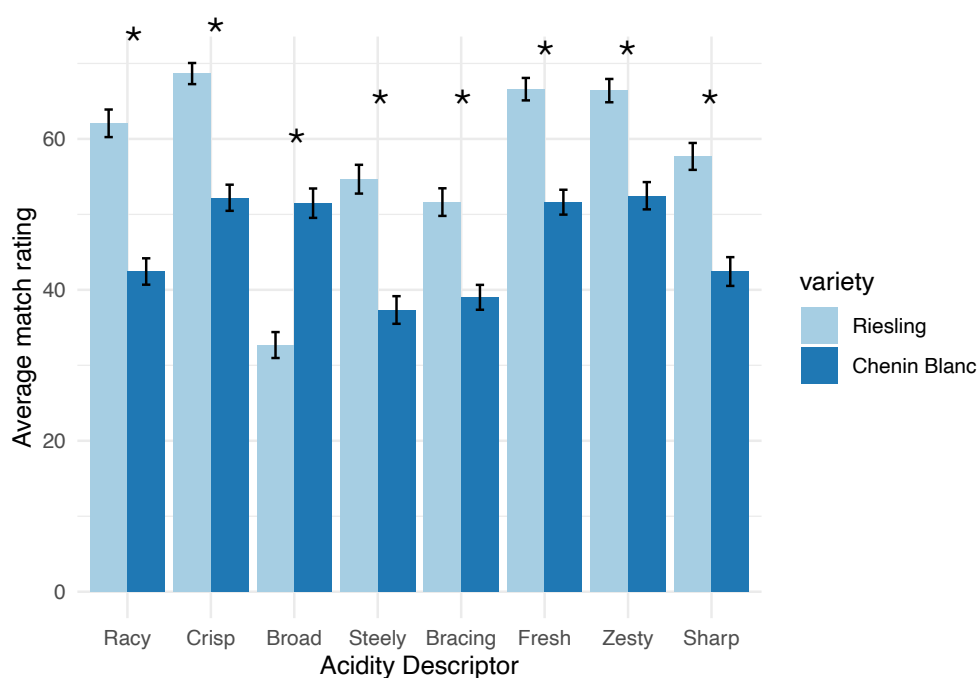


Figure 11. Average ratings of how well each descriptor matched each wine, grouped by variety. Error bars indicate standard error of the mean. Asterisks * indicate significant differences between varieties $p < .005$

To visualise the multivariate relationship between wine samples and descriptor ratings, a PCA was conducted (Fig. 12). Principal component 1 explains most of the variance (94.9%), with a clear differentiation between *broad* and all other descriptors. Principal component 2 explains only 3.4% of the variance, with more abstract descriptors such as *bracing* and *racy* on the positive end and more concrete fruit-related descriptors such as *zesty* on the negative end. The six wine types span the

space fairly evenly, with the exception of the two Marlborough wines being very close together, suggesting that experts gave them similar descriptor ratings despite the difference in grape variety. Otherwise, principal component 1 gives a clear separation of Rieslings (positive x-axis) versus Chenin Blancs (negative x-axis), suggesting that experts used more similar descriptors within varieties, and different descriptors across varieties. Along principal component 2, wines are divided by geographical origin, with wines from European origins on the positive y-axis (with the more abstract descriptors) and wines from non-European origins on the negative y-axis (with more concrete fruit-related descriptors), plausibly supporting a “old world” vs. “new world” differentiation.

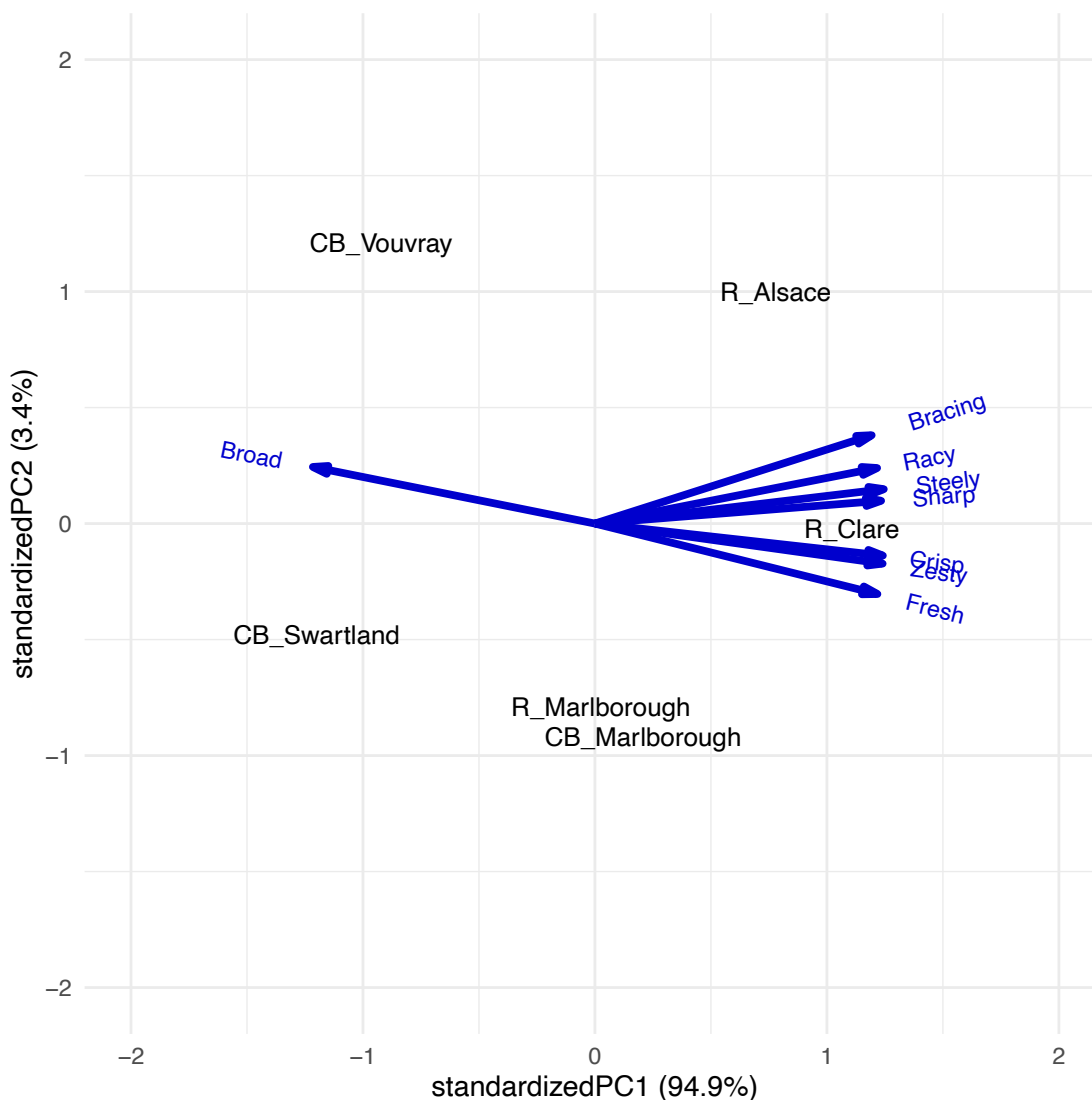


Figure 12. PCA biplot, with wines as scores and descriptors as loadings from the expert study.

4.2 RQ2

How is the wine's temporal acidity profile affected by grape variety and chemical composition?

4.2.1 Result of TI analysis

Based on the TI curves (Figure 5), analysis was conducted to compare the effect of grape variety (Riesling vs. Chenin Blanc) on AUC, $I_{\max}$, and $T_{\max}$. As shown in Figure 4 reposted below, AUC refers to the measured total perceived acidity intensity over time, $I_{\max}$ refers to the maximum intensity of acidity experienced over time, and $T_{\max}$ refers to the timepoint at which maximum intensity was perceived. Summary analyses of these metrics are shown in Table 3. ANOVA results revealed significant main effects for AUC ($F(1, 220) = 18.85, p < .001$), $I_{\max}$ ($F(1, 220) = 14.54, p < .001$), and $T_{\max}$ ($F(1, 220) = 22.62, p < .001$). In other words, there were significant differences between grape varieties in all three TI metrics. Post hoc analyses (Table 5) revealed the direction of these differences: Riesling wines had significantly higher AUC and $I_{\max}$ than Chenin Blanc, but Chenin Blanc wines had higher $T_{\max}$ than Riesling. In fact, it took on average 10 seconds longer for Chenin Blanc wines to reach peak acidity compared to Riesling.

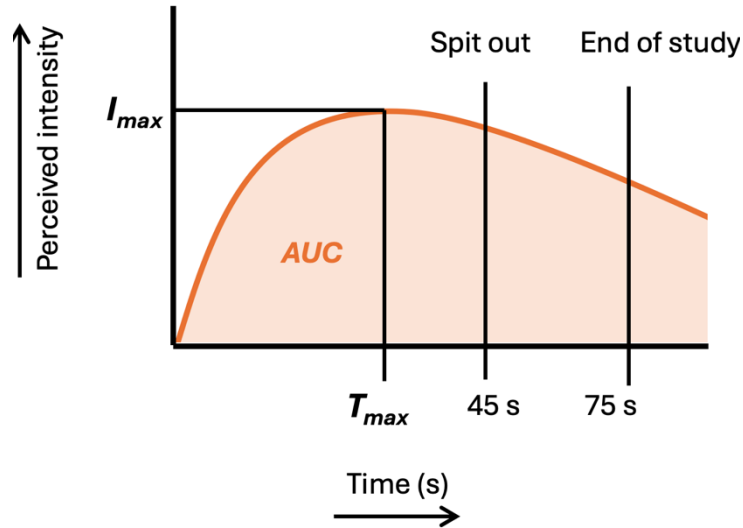


Figure 4. A TI curve with metrics of interest shown in bold italics, adapted from Goodstein et al., 2014. This is an idealised illustration not meant to reflect actual data. (Reproduced here from Section 3.3.3).

Table 5. Post hoc testing results for each TI metric, including adjusted means and standard deviations (SD) for each grape variety and p value.

	Chenin Blanc		Riesling		
	<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>	<i>p</i>
AUC	4261	19.6	4665	18.8	< .001
I_{max}	72.2	15.8	78.1	17.0	< .001
T_{max}	38.8	19.6	28.8	18.8	< .001

4.2.2 Relating TI with Chemical Analysis

Correlations between TI parameters and chemical parameters (Table 6) reveal that $I_{\max}$ has a significant negative correlation with pH. This is logical and validates the sensory panel performance, as it shows that wines with lower pH tend to have higher maximum perceived intensities (Plane et al., 1980). Supporting this finding, there was also a trending correlation between TA and $I_{\max}$, since wines with higher Total Acidity are also expected to have higher maximum perceived intensities. On the other hand, $T_{\max}$ is significantly correlated with ethanol, with higher ethanol levels corresponding

to greater $T_{\max}$. In other words, wines with higher ethanol levels tend to take a longer time to reach peak acidity intensity in the mouth.

Table 6. Pearson correlations (N=6) between TI metrics and chemical parameters A) directly related to acidity and B) not directly related to acidity. VA = volatile acidity. Citric, malic, tartaric, and lactic refer to acids. TA = total acidity expressed as tartaric acid. * Indicates $p < .1$, and ** indicates $p < .05$.

A)

	VA	Citric	Malic	Tartaric	Lactic	TA	pH
AUC	-0.08	0.28	-0.05	0.70	-0.55	* 0.75	* -0.81
Imax	-0.20	0.36	0.05	0.63	-0.57	* 0.79	** -0.81
Tmax	-0.06	-0.12	-0.31	-0.31	0.70	* -0.74	0.72

B)

	Fructose	Glucose	Glycerol	Reducing Sugar	Ethanol	CO2	Free SO2	Total SO2
AUC	-0.21	-0.39	-0.18	-0.23	0.09	0.30	0.21	0.19
Imax	-0.22	-0.22	-0.35	-0.22	-0.03	0.44	0.22	0.26
Tmax	-0.37	-0.54	0.65	-0.46	** 0.83	-0.28	0.25	0.35

4.3 RQ3

How does the relative relationship between chemical, perceptual, and psychological factors influence an expert's use of adjectives to describe acidity?

4.3.1 Predicting descriptor-matching scores

To probe the relationship between language use and chemical/sensory/psychological factors, partial least squared regressions (PLSR) were conducted to predict each descriptor's matching score based on each wine's chemical analysis, sensory evaluation, and tasting condition (nose clip vs. full retronasal olfaction). The results of these analyses are shown in Table 7, indicating key model parameters. It should be kept in mind that the PLSR models only explained between approximately 10-15% of the variance of the data. The remaining 85-90% of

data variance remained unexplained, which could be due to factors not included in the model (e.g., the wine's volatile composition, individual differences in sensory thresholds) or random variability.

Table 7. Indicators of PLSR model performance for each of the eight descriptors, arranged in order of decreasing model performance. The optimal component number for each model was derived from cross-validation. RMSE = root mean squared error, representing the average difference between model prediction and actual values.

Descriptor	Number components	% variance explained	RMSE
Crisp	3	15.4	24.67
Racy	3	14.8	28.24
Broad	2	14.5	28.87
Sharp	3	12.5	28.90
Fresh	3	12.7	24.80
Bracing	4	11.5	27.23
Steely	2	10.8	29.78
Zesty	1	9.5	26.77

To facilitate the interpretation of the models, Figure 13 illustrates the relationship between descriptor-matching scores and predictors using PLSR coefficient values. A visual examination of the figures shows that, across all 8 descriptors, the predictors with the strongest effects are those related to tasting condition, sensory analysis results (AUC and $I_{\max}$), and overall measures of acidity (pH and TA).

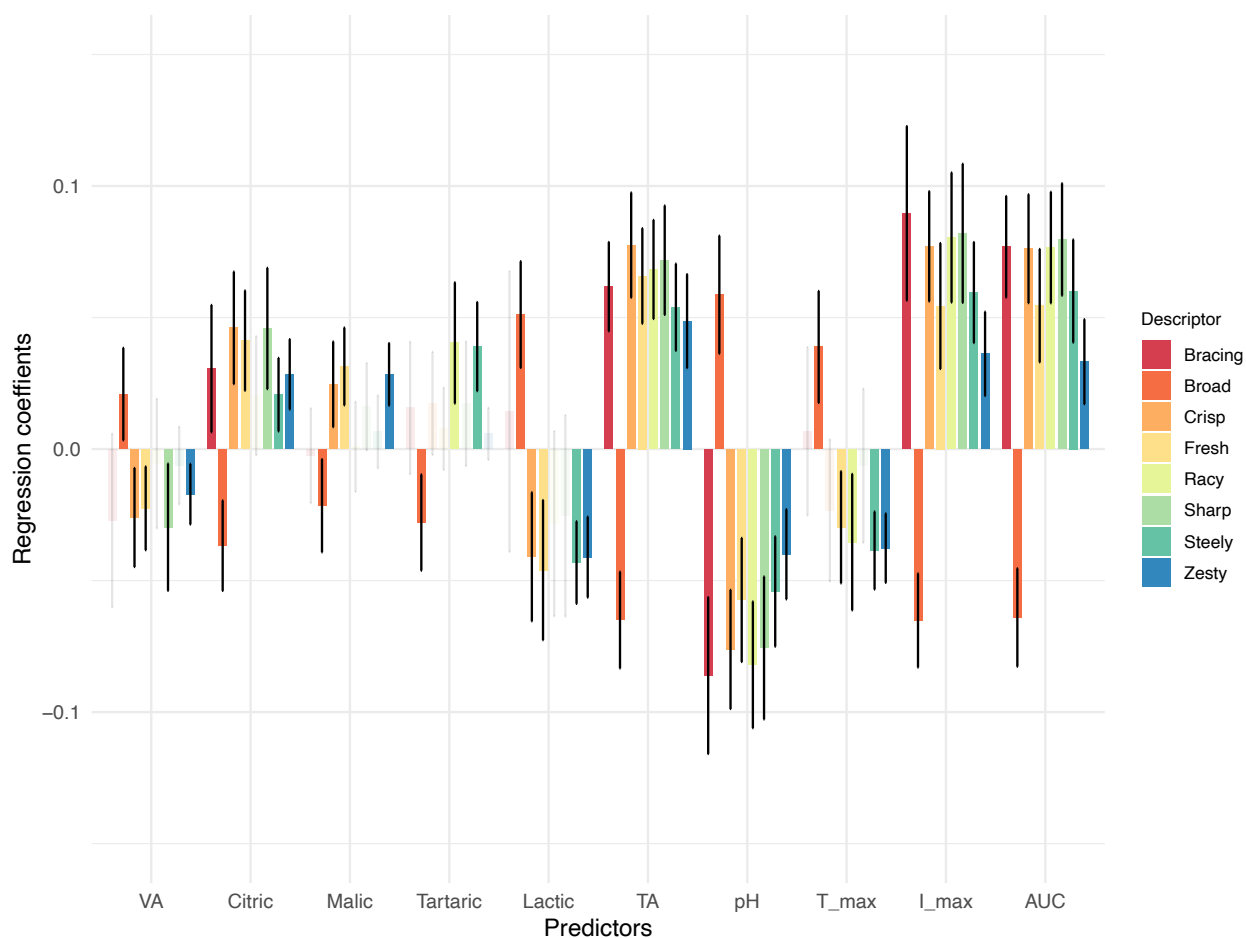


Figure 13A Calculated regression coefficients using PLSR, illustrating, for each of the eight acidity descriptors, which predictors best correlate with descriptor-matching scores. Predictors correspond to sensory analysis results and chemical properties directly related to acidity. Error bars indicate 95% confidence intervals. Solid colouring indicates statistically significant coefficient values.

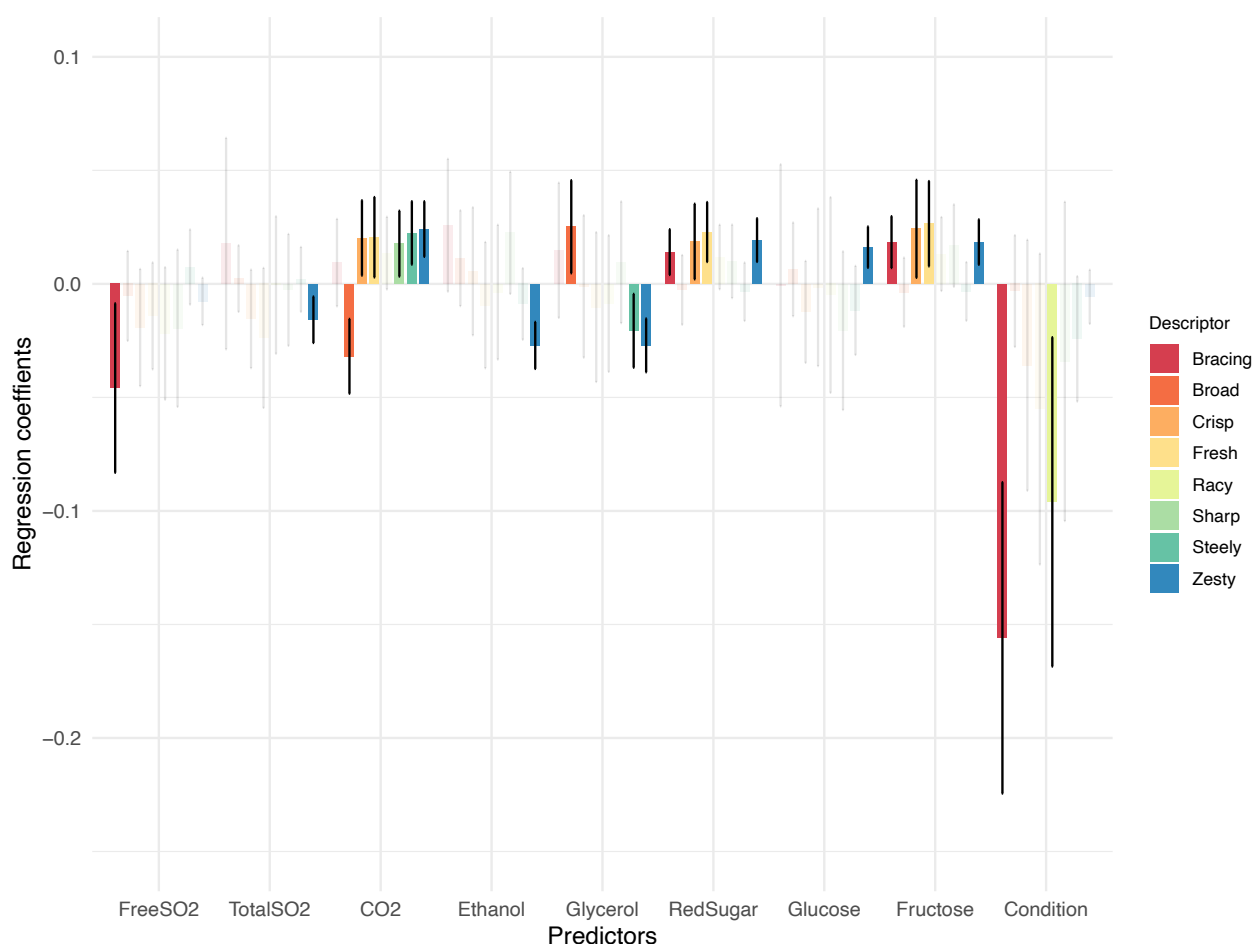


Figure 13B. Calculated regression coefficients using PLSR, illustrating, for each of the eight acidity descriptors, which predictors best correlate with descriptor-matching scores. Predictors include tasting condition (negative value = no nose clip) and non-acid chemical compounds. Error bars indicate 95% confidence intervals. Solid colouring indicates statistically significant coefficient values.

Tasting condition was a significant predictor for the descriptors *bracing* and *racy*, where not wearing nose clips led to higher matching ratings for those descriptors (Fig. 13A), suggesting that these descriptors are not purely gustatory but reflect aroma-taste associations built through experience. In terms of sensory analysis predictors, AUC and $I_{\max}$ were positively associated with *sharp*, *racy*, *crisp*, and *bracing*; and to a lesser extent with *zesty*, *steely*, and *fresh*; and negatively associated with *broad*. $T_{\max}$ on the other hand, was negatively associated with *zesty*, *steely*, *racy*, and *fresh*.

In terms of chemical composition, higher TA and lower pH were both associated with higher matching scores for all acidity descriptors except for *broad* (Fig. 13B), a pattern also reflected by citric acid despite its overall low levels (Table 2A). Lactic acid negatively correlated with *zesty*, *steely*, *fresh*, and *crisp*, but positively with *broad*. Tartaric acid showed positive correlations with *racy* and *steely*, while malic acid correlated positively with *zesty*, *fresh*, and *crisp*, but negatively with *broad*.

Non-acid-related chemical components also influenced the choice of descriptors, albeit to a lesser extent (Fig. 13B). Residual sugar (driven by fructose, see Table 2B) was positively correlated with *zesty*, *fresh*, *crisp*, and *bracing*. Glycerol was negatively correlated with *zesty* and *steely*. Ethanol was positively correlated with *bracing* and negatively correlated with *zesty*. CO₂ was also a highly significant predictor, being positively associated with *zesty*, *steely*, *sharp*, *fresh*, *crisp*, and negatively associated with *broad*. Free SO₂ level is negatively correlated with *bracing*.

4.3.2 Visualising language-chemistry-sensory relationships

To visualise the multivariate relationships between descriptors, chemical, and sensory data, a PCA was conducted (Fig. 14). Principal component 1 explains 30% of the variance, driven by TA on the one hand and lactic acid on the other. There is a clear differentiation between *broad* along the negative dimension and all other descriptors on the positive dimension. Similar to PLSR results above (Fig. 13A), *broad* has positive correlations with lactic acid and glycerol. On the other hand, *zesty* and *fresh* are most correlated with malic acid and TA, whereas *sharp* and *steely* are most closely correlated with citric acid and CO₂. The wines are also clearly separated along principal component 1, with Rieslings located along the positive direction and Chenin Blancs (mostly) along the negative direction. The Chenin Blanc from the third “non-classical” region, Marlborough, also sits along the positive direction, suggesting that it

may have more in common with the Rieslings in terms of acidity composition (citric, malic, and TA) and descriptor use.

Principal component 2 explains 21% of the variance, with tartaric acid and SO₂ along the positive end, and sugar compounds on the negative end. The descriptors don't have much separation along this component, with a slight tendency of *broad*, *zesty*, and *fresh* along the negative direction (similar to PLSR findings with these descriptors' association with wines containing slightly higher levels of residual sugar).

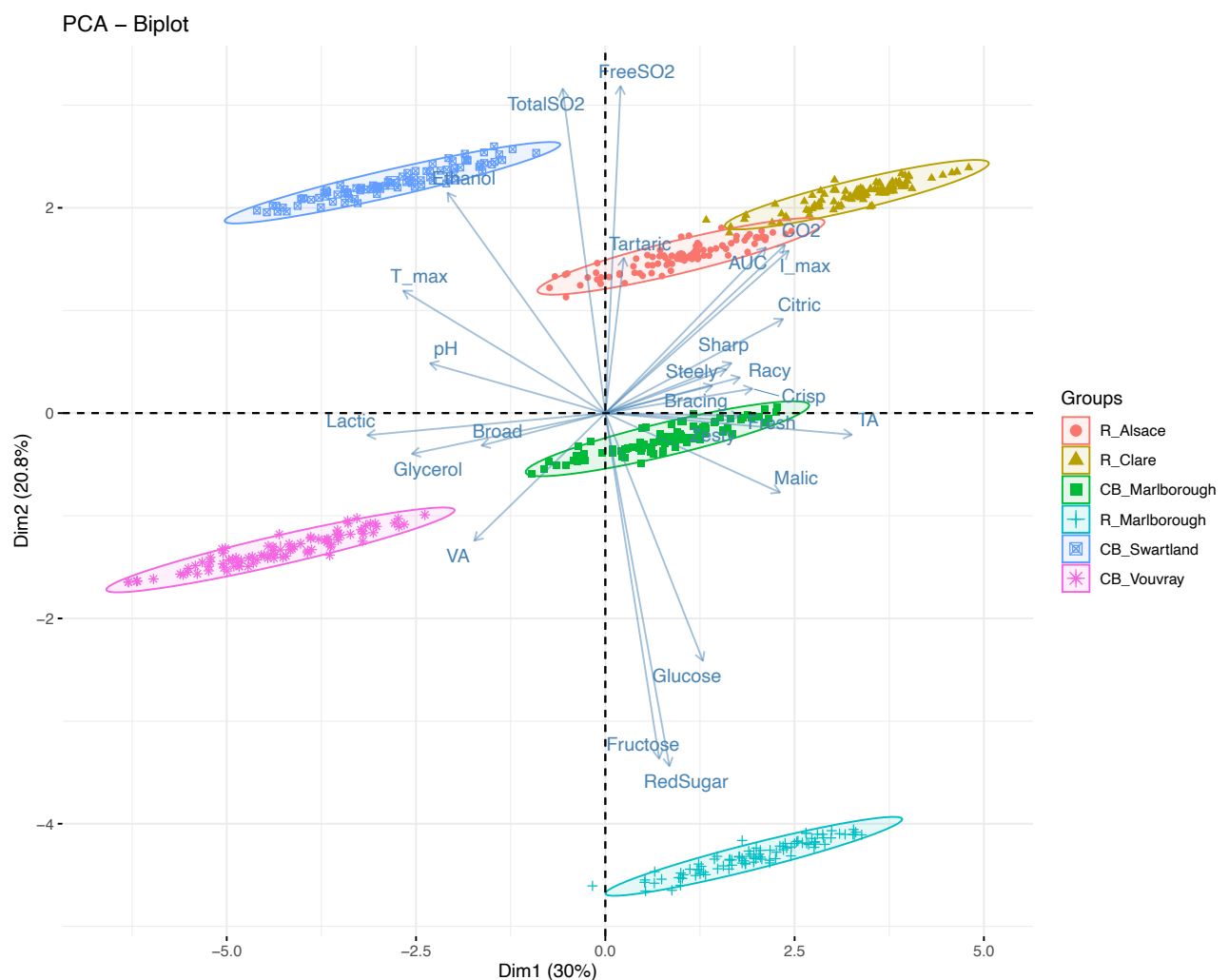


Figure 14. Biplot of PCA with expert descriptor match-ratings as scores, colour-coded by wine type with confidence ellipses. Loadings consist of acidity descriptors, chemical analysis, and sensory analysis to visualise the relationship between them.

5. Discussion

This section explores the implications of the findings from each research question, analysing how they contribute to both theoretical understanding of wine cognition and practical applications in tasting and education.

5.1 RQ1

How do wine experts use adjectives to describe the quality of acidity in Riesling and Chenin Blanc wines, and are there consistent variety-based differences?

In summary, the findings from section 4.1 demonstrate clear variety-based differences in how wine experts describe the quality of acidity in Riesling and Chenin Blanc. Results from the Online Questionnaire revealed that descriptors such as *zesty*, *racy*, and *steely* were more frequently associated with Riesling, while *bracing* and *crisp* were more commonly used for Chenin Blanc (Table 4). As seen from the Correspondence Analysis (Fig. 10), experts from the online questionnaires tended to use different acidity descriptors with each of the four representative wine styles, showing that even within the same variety, different origins can result in different descriptor tendencies. Results from the Expert Tasting partially supported Online Questionnaire findings in that different acidity descriptors were more often associated with specific varieties in a blind tasting setup. In this case, *broad* was more likely used with Chenin Blanc and all other descriptors (*bracing*, *racy*, *steely*, *sharp*, *crisp*, *zesty*, *fresh*) with Riesling (Fig. 11).

5.1.1 Theoretical Implications

Based on the online questionnaire, the consistent use of descriptors like *zesty* and *racy* for Riesling, and *broad* for Chenin Blanc, suggests that wine experts draw upon stable, variety-specific language to describe acidity. This consistency reinforces the idea that certain perceptual qualities of acidity are closely tied to the chemical and

sensory properties of the grape variety. These findings align with previous research on crossmodal correspondences, where shape, colour, texture, and sound are associated with taste, suggesting that tasters use metaphorical associations to articulate complex sensory experiences (Spence et al., 2013; Wang et al., 2024).

The difference between the clear separation of descriptors between wine styles in the online questionnaire (Fig. 10), versus the similarity in actual usage during blind tasting (Fig. 12), highlights a difference between conceptual representation and perceptual ratings of acidity descriptors. While online experts had a clear mental separation of the stereotypical wine styles in the descriptor space, when it came to the in-person tasting, the choice of acidity descriptor usage was much more simplified. In cognitive terms, this reveals a gap between schema-driven expectations and sensory-driven decisions (Parr et al., 2002; Spence & Wang, 2019). This discrepancy speaks to a broader challenge in wine cognition: *mental models are neat, but perception is messy*. Blind tasting, therefore, exposes the instability of our sensory judgments when detached from context. This is precisely why blind tasting is so difficult, not just because perception is complex, but because our internal expectations often outpace what the glass actually delivers.

5.1.2 Practical Implications

The practical implications of these findings are particularly relevant for wine education and communication. Based on the online questionnaire, the consistent use of acidity descriptors for specific varieties, such as *zesty* and *racy* for Riesling and *broad* for Chenin Blanc (Table 4), suggests that those involved in the MW program already have some preconceived ideas for style-specific acidity descriptors. That said, Correspondence Analysis (Fig. 10) also showed that other descriptors, such as *piercing*, *bright*, and *firm*, seem to be used equally likely for all four more established

wine styles (Alsace and Clare Valley Riesling, Vouvray and Swartland Chenin Blanc). This suggests a nuanced preconception of acidity descriptors, with some acidity descriptors being variety or style-specific, while others are used for a wider range of wines. When comparing the online questionnaire with actual expert tasting results, there was a difference in the distribution of descriptor usage. For example, while bracing was more associated with Chenin Blanc in the online questionnaire, it was associated more with Riesling in the blind tasting. This points to the difference between mental constructs of wine styles and real-world experience. After all, not all wines in real life taste like idealised mental representations. This suggests that the participants were in fact *blind-tasting* and describing their sensations of acidity, rather than guessing the grape varieties and then applying the associated descriptors.

One of the key challenges in teaching and evaluating acidity is the lack of a sensory reference standard for metaphorical descriptors. Unlike measurable traits like pH or residual sugar, terms such as *racy* or *steely* have no fixed calibration; who is to say what level or quality of acidity qualifies as *steely* or *racy*? This ambiguity can lead to inconsistency in how such terms are used. A related risk is that tasters may rely too heavily on variety-specific expectations: in blind tastings, tasters may identify a wine first, then retroactively apply the “appropriate” acidity descriptors they associate with that grape or region. This reverse reasoning prioritises identification over genuine sensory analysis and may obscure how acidity is actually perceived in the moment. To support more consistent and perceptually grounded language use, tasting education should go beyond rote memorisation of “typical” descriptors. Structured training exercises could instead isolate and manipulate specific elements in a mixture, such as residual sugar, ethanol, or acid composition, allowing students to experience firsthand how these variables modulate acidity perception, regardless of grape variety.

5.2 RQ2

How is the wine's temporal acidity profile affected by grape variety and chemical composition?

Results from section 4.2.1 revealed significant differences in the temporal profiles of acidity between Riesling and Chenin Blanc, suggesting a “crescendo” description of Chenin Blanc acidity, which takes a longer time to arrive at maximum intensity. However, correlational analysis (Table 6) offers a different interpretation, pointing to a significant positive association between ethanol level and $T_{\max}$. This suggests that, rather than variety-specific differences, higher ethanol levels may delay the perception of peak acidity. In other words, temporal aspects of acidity perception are not only influenced by grape variety but also closely tied to the wine's chemical composition.

5.2.1 Theoretical Implications

Riesling's faster rise to peak acidity and higher intensity (Table 3) suggest a more immediate, sharper sensation of acidity, which may explain its frequent association with descriptors like *racy* and *crisp*. In contrast, Chenin Blanc's slower build-up to peak intensity aligns with descriptions like *broad*, reinforcing the idea of an extended, more gradual rise in acidity (Table 3). Taken at face value, these differences support the notion of “acid shape” proposed by Nick Jackson MW, where Riesling is often described as having a vertical or immediate acid profile, while Chenin Blanc is perceived as having a more crescendo-like evolution (Jackson, 2022). However, statistical analysis section 4.2 indicates that these differences may not be inherently variety-driven. Chenin Blanc wines in this study had, on average, higher ethanol levels than the Rieslings (Table 2B), which may account for the delayed perception of acidity. A further argument against variety-specificity is that Riesling and Chenin Blanc wines

from Marlborough received similar acid descriptor ratings, while the other four “classic” wines were more distinct (Fig. 12). The inclusion of this third pair of Marlborough wines therefore proved valuable in revealing regional or producer influences may exert a greater impact on perceived acidity than grape variety alone.

The significant correlation between ethanol content and time to maximum intensity ($T_{\max}$) raises important questions about how chemical interactions in the mouth, particularly the masking effect of ethanol, may delay the perception of acidity. This aligns with recent research suggesting that higher ethanol concentrations can reduce the intensity of sourness (Thibodeau & Pickering, 2021), providing a possible explanation for the slower build-up of acidity in Chenin Blanc wines with higher ethanol levels. In other words, it remains to be seen whether perceived acidity evolution reflects variety-based factors (Jackson, 2022) or chemical composition.

5.2.2 Practical Implications

As these results suggest, students learning to evaluate wine should focus not only on the intensity of acidity, but also on how acidity evolves over time. Incorporating temporal elements into sensory training can help students enhance their ability to describe and identify wines more accurately, for example, during exams and professional tastings. In particular, educators could introduce a “15-second rule” tasting protocol by asking students to evaluate acidity twice, once immediately upon tasting, and once after 15 seconds, to train awareness of dynamic acidity evolution in the mouth. As Figure 5 shows, the initial increase in acidity takes place within the first 15 seconds. This exercise would lead to a more nuanced assessment capturing both attack and mid-palate, and help students improve tasting consistency by standardising sip-hold-spit behaviour. This aligns with standard practices from Sensory Science (Lawless & Heymann, 2010), where panellists are routinely given very specific eating

instructions, and would help avoid the variability introduced by inconsistent mouth movements.

Another important finding was the role of ethanol in delaying perceived peak acidity ($T_{\max}$). This has practical relevance for both winemakers and tasters, as it highlights the need to consider ethanol as a factor when assessing the balance of a wine's structure. Wines with higher alcohol may require tasters to wait longer to fully perceive the acidity. This reinforces the importance of consciously extending evaluation time, especially in professional settings, to avoid prematurely judging a wine as flabby. For winemakers, this means that achieving freshness in high-alcohol wines may require elevated acidity or structural counterpoints such as CO₂ or bitterness.

5.3 RQ3

How does the relative relationship between chemical, perceptual, and psychological factors influence an expert's use of adjectives to describe acidity?

The results from section 4.3 reveal that the relationship between chemical composition, sensory perception, and the use of acidity descriptors is complex, with only a portion of the variance in descriptor-matching scores explained by chemical and sensory factors. Multivariate regression analysis (Fig. 13) revealed that the choice of acidity descriptors is significantly correlated with both different types of acidity as well as non-acid components, temporal evolution, and aroma availability.

5.3.1 Theoretical Implications

First, significant regression correlations in the models do support the fact that language use is indeed shaped by the wine's chemical composition. For instance, *zesty* and *fresh* are significantly associated with higher levels of malic acid and lower levels of lactic acid. Moreover, the involvement of non-acidic compounds, such as

ethanol and CO₂, in shaping descriptors like *zesty*, *steely*, and *fresh* challenges the traditional view that acidity descriptors are solely driven by acid content. These findings suggest that wine experts rely on a broader set of chemical cues when describing acidity, reinforcing the idea that cognitive representations of wine acidity are influenced by multiple interacting sensory and chemical factors.

Secondly, these results reveal some intriguing parallels and contrasts with recent findings regarding the associations between specific acids and acidity descriptors (Honoré-Chedozeau et al., 2024). In both studies, malic acid was consistently linked to descriptors that evoke freshness; in the present study, *fresh* was strongly correlated with malic acid levels, and Honoré-Chedozeau similarly found *apple* and *freshness* associated with malic acid. Similarly, tartaric acid is associated with a metallic character. In this study, *steely* and *racy* were most closely associated with tartaric acid, while Honoré-Chedozeau found tartaric acid more frequently described as *astringent* and *metallic*. Finally, lactic acid is associated with a lack of angularity or sharpness, with significant correlations with *broad* in this study and *roundness* by Honoré-Chedozeau. Overall, the results of both studies confirm the idea that different acid types have inherently different taste expressions, which can lead to a consistent use of metaphoric language.

The role of aroma in shaping how tasters describe acidity is an important theoretical result from this research. Evidence from the blind tasting conditions comparing noseclip versus full retronasal access showed that the absence of aroma significantly reduced the use of descriptors *bracing* and *racy*. This suggests that these descriptors are not purely based on gustatory or trigeminal cues, but may reflect learned aroma–acidity associations. In other words, tasters may choose certain acidity descriptors not in response to the taste itself, but because the aroma primes them to

expect a particular taste sensation. This finding reinforces prior research on multisensory integration, showing that flavour perception is a cognitive construct where aroma, taste, and trigeminal inputs converge (Spence et al., 2013). It also suggests that acidity descriptors may be less objective than they appear, being shaped not just by the wine's chemical composition, but by the associative memory of the taster.

That said, the relatively low predictive power of the regression models, which explained only about 10-15% of the variance, suggests that acidity descriptors are influenced by more than just the chemical, sensory, and psychological factors included in the models. This implies that other chemical components, such as the aroma/volatile components, as well as perceptual and psychological factors, including cognitive associations and individual tasting experiences, likely play a significant role in how descriptors are selected, supporting the idea that wine communication is inherently multisensory and subjective.

5.3.2 Practical Implications

The findings from section 4.3 have significant implications for how people assess and communicate acidity in wine. For wine education, the fact that different organic acids are associated with different descriptors (Fig. 13A), such as malic acid with *fresh*, suggests targeted training to familiarise students with their individual qualities. For example, students can taste reference solutions of different acids at different intensity levels and learn to generate descriptors that reflect each acid's unique profile. This would train not only sensory sensitivity but also the ability to use language to describe their sensory experiences, which is crucial in both blind tasting and professional communication (Croijmans et al., 2024). Moreover, results showing the importance of non-acid compounds in acidity description (Fig. 13B) suggest that

students should be taught to consider the full sensory experience when evaluating acidity. For example, educators can set up tasting exercises where the same wine is compared before or after carbonation, or with varying levels of sugar or aroma added.

From a technical perspective, these findings challenge the assumption that acidity descriptors can be accurately inferred from chemical data alone. This means that lab-reported values such as TA and pH, while informative, may not be sufficient predictors of perceived acidity. Winemakers, for example, should therefore validate acidity levels through human sensory evaluation, rather than relying solely on analytical results.

5.4 Limitations and future work

While this research has provided valuable insights into the relationship between acidity perception, chemical composition, and (English) language use in wine tasting, several limitations should be acknowledged. First, the study was restricted to a relatively small sample of six wines, which limits the generalisability of the findings. Furthermore, there were minor setbacks in terms of wine acquisition, such that the full intended wine selection criteria (Section 3.1.1) could not be met in full. For example, while there was an attempt to control for vintage variation, the two wines from Marlborough are from 2023, while the rest of the wines are from 2022. This was due to very limited supplier availability (lack of producers who made both Riesling and Chenin Blanc from the same region and in the same style) and constraints of a tight purchasing deadline. Additionally, chemical analysis (Table 2) revealed that not all the wines met the intended criteria of having strictly <5 g/L residual sugar. Specifically, Reducing Sugar measures were around 6 g/L for the Vouvray and Marlborough Chenin Blanc, and 10 g/L for the Marlborough Riesling. This discrepancy was discovered only after the chemical analysis was conducted and the wines had already been sourced.

Since Total Acidity in all cases was no more than 2 g/L lower than the sugar content, the rest of the studies were conducted under the assumption that the wines were perceptually dry (Noordeloos & Nagel, 1972). Future studies could therefore benefit from a larger dataset with a broader range of wines, including different grape varieties, regions, and winemaking styles.

In terms of population sampling, the majority of wine experts from the Online Questionnaire and Expert Tasting were MW students and MWs, which introduces potential bias. These participants may have been more familiar with certain acidity descriptors due to their training, possibly limiting the findings to a specific cohort of English-speaking wine professionals. That said, the decision to focus on MWs and MW students, especially for the Online Questionnaire, was due to their increased familiarity with using acidity descriptors. The results of this project are intended to support MW students in refining their use of acidity descriptors to communicate with greater precision and consistency.

In addition, participant selection for the expert tasting study was limited to those who could physically come to the study in Copenhagen or London, which limited geographical representation. While the online questionnaire surveyed MWs and MW students, the expert tasting study recruited those who have WSET Diploma certifications, who may or may not have been involved in the MW programme. Ideally, both the online questionnaire and expert tasting study would have drawn from the same population sample, but this was not possible due to logistical constraints and sample size requirements. In the expert tasting, therefore, wine education level and familiarity with specific terms may have influenced the choice of descriptors. As shown in Figure 8, participants were relatively less familiar with terms such as *racy*, *broad*, and *bracing*. The eight descriptors used in the expert study were drawn from the most

common terms in the online questionnaire, suggesting a possible discrepancy in language use between WSET Diploma holders involved in the MW programme and those who are not. As the MW programme encourages the use of acidity descriptors, some terms familiar to MWs and MW students may be less familiar to others.

Another potential limitation was the experimental design for the expert tasting, where all participants first wore nose clips before moving to the full retronasal condition. This sequential approach could have introduced learning effects, as participants may have developed expectations about the wines during the first round of tastings. Future studies could counterbalance the order of tasting conditions or introduce more diverse sensory conditions to reduce this potential bias.

Looking ahead, future work should consider including more detailed chemical analyses, such as gas chromatography-mass spectrometry (GC-MS), to identify specific flavour compounds that influence acidity perception. This could help explain the relatively low predictive power of the models used in this study, as certain key aroma compounds might not have been accounted for. Exploring how these compounds interact with acidity could shed light on the olfactory contributions to the perception of acidity.

6. Conclusions

This research challenges conventional thinking about how acidity is perceived, described, and communicated in wine. Results demonstrated that acidity is not a fixed structural trait, but a multisensory and dynamic experience shaped by time, chemical composition, aroma integration, and the cognitive lens through which tasters interpret the wine. Descriptors like *racy* and *broad* were shown to vary systematically across grape varieties, while the temporal evolution of acidity and the influence of matrix components, such as ethanol and carbonation, further influenced how acidity is described. Furthermore, while expert tasters demonstrated clear mental constructs linking wine styles to specific acidity descriptors in the online survey, these distinctions largely collapsed during blind tasting, where descriptor use became more perceptually grounded and less driven by stylistic expectations.

Tasting education must therefore move beyond static definitions of acidity and train tasters to observe how acidity unfolds over time. Teaching exercises should include holding the wine in the mouth, noting the timing of peak acidity, and using standardised mouth movements to ensure consistent evaluations. Exercises using single-acid solutions (e.g., malic, tartaric, lactic), comparisons of wines with altered single components (e.g., sweetness, carbonation), and structured timing protocols (e.g., “15-second rule”) can help build both sensory accuracy and linguistic consistency.

Ultimately, this research calls for a more perceptually grounded, temporally aware, and chemically informed approach to wine tasting. Rather than treating acidity as a single metric, we must engage with it as a time-based, multisensory phenomenon. Doing so is essential for improving the quality of tasting education, the precision of

professional language, and the reliability of how wine is judged, described, and understood.

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Appendices

Appendix A. Accepted version of RPP

IMW Research Paper Proposal Submission Form			
Student ID	28515	Date of submission	9 October 2024
RPP Version No	6	Name of Advisor	Stephen Wong
Note: RPPs must be submitted via your Advisor to the IMW			
Proposed Title			
<i>Communicating acidity: Investigating the relationship between language use, chemical composition, and temporal sensory perception of Riesling and Chenin Blanc</i>			
Research Questions: Define the subject of your Research Paper and specify the specific research questions you plan to pursue. (No more than 300 words)			
While acidity is a key structural element in wine, there has been little research to investigate the way people use metaphorical language to describe the quality of acidity (e.g., <i>broad</i>, <i>steely</i>). This study will investigate both <i>how</i> experts use acidity descriptors and to be able to predict <i>why</i> specific descriptors are used. Research Questions: RQ1: How do wine experts use adjectives to describe the quality of acidity in Riesling and Chenin Blanc wines, and are there consistent variety-based differences? RQ2: How is the wine's temporal acidity profile affected by grape variety and chemical composition? RQ3: How does the relative relationship between chemical, perceptual, and psychological factors influence an expert's use of adjectives to describe acidity?			
Background and Context: Explain what is currently known about the topic and address why this topic requires/offers opportunities for further research. (No more than 300 words)			

Acidity descriptors

Acidity is a key structural element in wine. There are different ways that tasters are taught to assess acidity. WSET students evaluate the level of acidity (low to high), while both level and quality of acidity should be stated for the MW exam. The quality of acidity is often conveyed via multisensory metaphorical descriptors referring to vision (e.g., bright), space (e.g., broad), shape (e.g., sharp), or touch (e.g., hard). For example, Tim Wildman MW teaches that Riesling acidity tends to be crisp, racy, and steely, while Chenin Blanc tends to have broad, firm, and tart acidity (winetutor.tv). Nick Jackson MW in his book “Beyond Flavour” (2020) takes this one step further by proposing the concept of *acid shape* – the temporal evolution of acidity levels in a single sip from the attack to the finish. According to Jackson, each grape variety has signature shapes that remain constant regardless of origin.

Despite these common educational references, there has been no scientific research to support or explain any consistent use of acidity descriptors for different grape varieties. There is, however, a wide body of literature on the use of multisensory metaphors or crossmodal correspondences that have been consistently applied to describe wine (e.g., Crichton-Fock et al., 2023).

Acid composition in grapes

While the factors behind the composition of different organic acids in grapes are complex, ranging from climatic influences to shoot vigor (Jackson 2020), hereditary factors can affect grape acid content (Kliewer et al., 1967). For example, while Riesling and Chenin are both high acid varieties, Riesling is inherently higher in tartaric acid whereas Chenin Blanc is proportionally higher in malic acid (Kliewer et al., 1967). Therefore, it is possible that any differences in variety-specific acidity descriptors could feasibly come from differences in acid composition.

Acidity perception

Acidity perception is the sum of contributions from the wine and the taster. Wine contains a matrix of different organic acids which is determined by grape variety as well as harvest ripeness and winemaking methods. From the perspective of the taster, acidity is due to a combination of physiological (e.g. sour taste receptors, salivation) and psychological (e.g., taste-taste and aroma-taste interactions) factors. In sensory science, the intensity of acidity in food and drink can be measured either by its absolute intensity or by its temporal evolution in the mouth using Time Intensity (e.g., Goodstein et al., 2014).

Put together, this study aims to investigate factors behind wine experts' use of acidity descriptors in Riesling and Chenin Blanc, two of the most common high-acid white grape varieties, using both instrumental/analytical and psychological/sensory methods.

Sources: List key background sources (previous research, articles, books, official documents etc) that will help to form your literature review. Note these should be listed as full references. (No more than 200 words)

Language of acidity descriptions:

- Jackson, N. (2022). *Beyond flavour: Wine tasting by structure*. UK: Amazon.
- *Tasting Vocabulary*. Winetutor.tv. <https://winetutor.tv/exam-technique/technique-practical/>
- Crichton-Fock, A., Spence, C., & Pettersson, N. (2023). Using crossmodal correspondences as a tool in wine communication. *Frontiers in Psychology*, 14, 1190364. <https://doi.org/10.3389/fpsyg.2023.1190364>

Acidity in wine:

- Jackson, R. S. (2020). *Wine Science: Principles and Applications, Fifth Edition*. Cambridge, Massachusetts: Academic Press.

- Kliewer, W. M., Howarth, L., & Omori, M. (1967). Concentrations of tartaric and malic acids and their salts in *Vitis vinifera* grapes. *American Journal of Enology and Viticulture*, 18, 42-54.

Physiology of acidity perception:

- Smith, D. V., & Margolskee, R. F. (2001). Making sense of taste. *Scientific American*, 284, 32-39.
- Lawless, H. T., & Heymann, H. (2010). *Sensory evaluation of food: Principles and practices, Second Edition*. New York, NY: Springer.

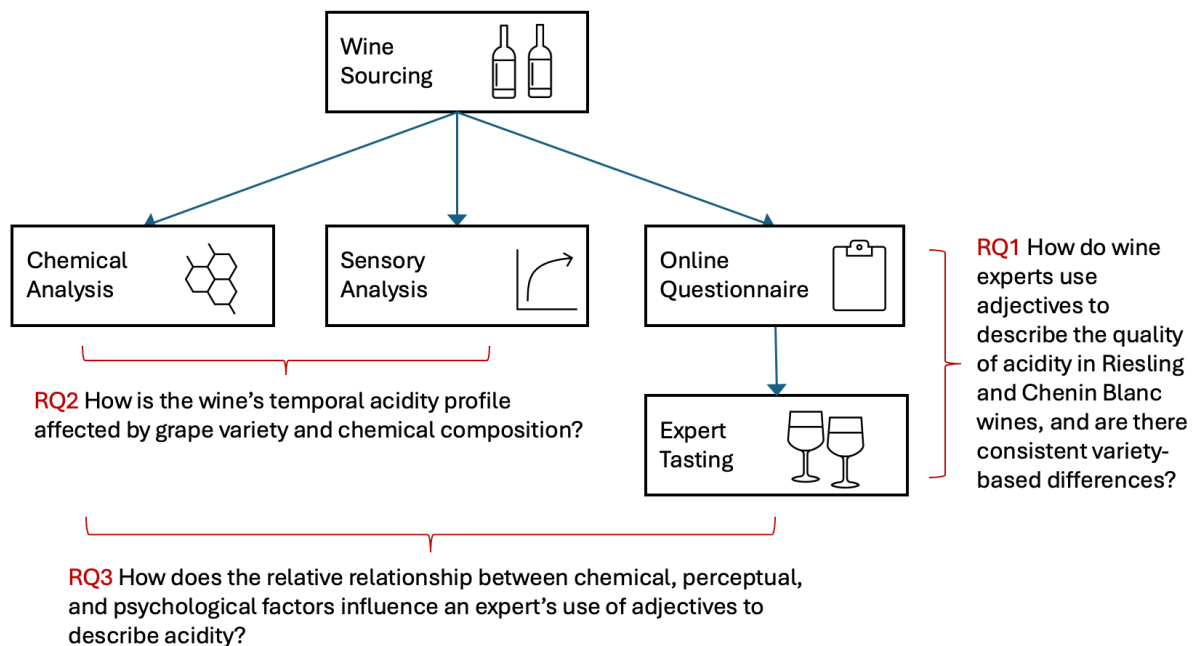
Measuring sensory perception:

- Heymann, H., & Ebeler, S. (2017). *Sensory and Instrumental Evaluation of Alcoholic Beverages*. Cambridge, Massachusetts: Academic Press.
- Peyvieux, C., & Dijksterhuis, G. (2001). Training a sensory panel for TI: A case study. *Food Quality and Preference*, 12, 19–28.
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<https://doi.org/10.1145/3141788.3141792>

Research Methodology: Detail how you will identify and gather the material or information necessary to answer the research question(s) and discuss what techniques you will use to analyse this information. (No more than 800 words)

The research project will be conducted via six interconnected work packages (WP) that target the three research questions:



WP1: Wine selection and sourcing

- High acid varieties are chosen to focus on acid *quality* (thus controlling for acid *level*), Riesling and Chenin Blanc (CB) are selected because they are major international varieties, have different acid compositions (Kliwer et al. 1967) and distinctive “acid shapes” (vertical for Riesling, crescendo for CB, Jackson, 2022). While all styles are made with these varieties, the study will focus on dry still wines (residual sugar < 5 g/L) to avoid acidity interference effects from sweetness or carbonation (Heymann & Ebeler, 2017).
- For each variety, the two most representative origins (i.e., most frequent use in last 15 years' MW exams) are selected, plus a third region with a single producer making both varieties in a similar way (e.g., Quails' Gate in Okanagan or Astrolabe in Marlborough) to help control for geographical and winemaking differences. 6 samples in all.

- Additional selection criteria to ensure comparability: most recent vintage, no MLF, no new oak, 12-13% abv. Producers used in previous MW exams will be prioritized depending on availability in both Denmark and UK⁷.

	Riesling	CB
Region 1	Clare Valley	Vouvray
Region 2	Alsace	Stellenbosch
Region 3 (same producer)	Okanagan / Marlborough	Okanagan / Marlborough

WP2: Chemical analysis

- Wines: 6 samples from WP1 in duplicates (12 samples in all)
- Procedure: Samples will be analysed by FOSS (Hillerød, Denmark) via WineScan™ or at ISO-certified laboratory.
- Metrics: Including but not exclusive to: Ethanol, Fructose, Glucose, Glycerol, pH, Total Acidity, Malic Acid, Lactic Acid, Tartaric acid, Volatile Acidity

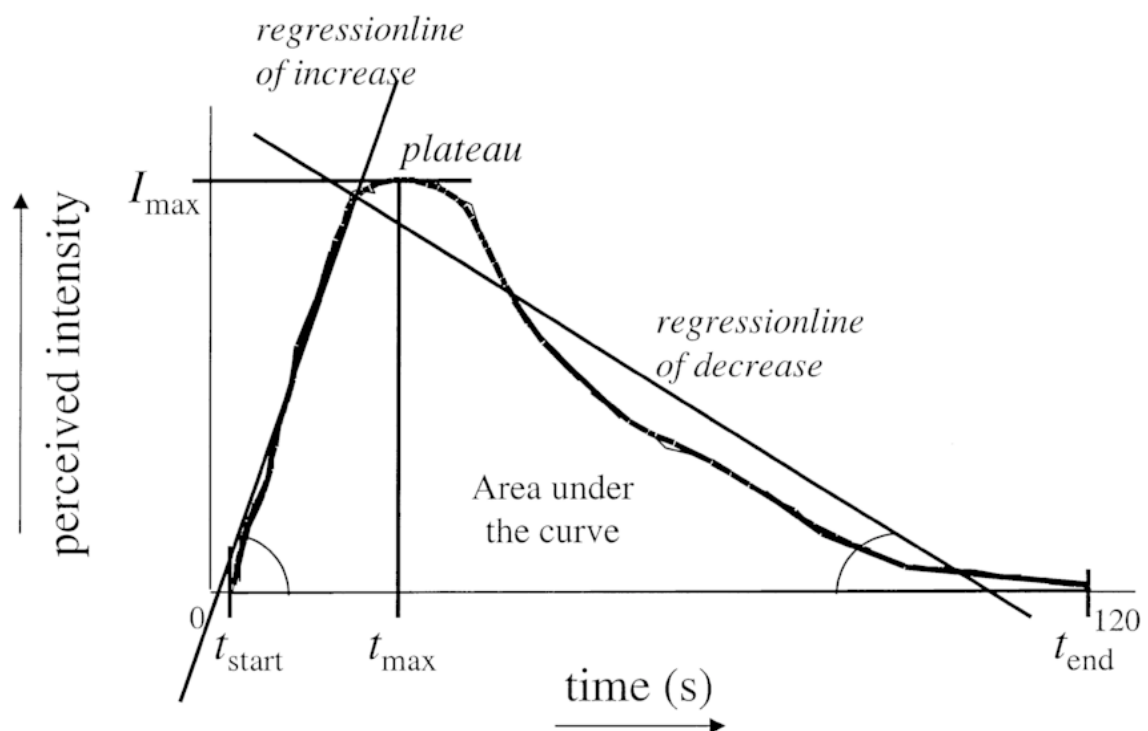
WP3: Sensory analysis by trained panel

- Location: University of Copenhagen Dept of Food Science (UCPH Food) sensory lab
- Participants: 8-12 panelists (Heymann & Ebeler, 2017) will be recruited from students/staff at UCPH Food
- Wines: 6 samples from WP1, served blind in ISO glasses in 25 mL portions
- Software platform: Compusense
- Procedure: Participants will come for three 1-1.5 hour sessions to be trained on the time intensity (TI) method whereby participants will hold a mouthful of wine for approximately 30 seconds and evaluate acidity intensity at each point in time (Goodstein et al., 2014).

⁷ Availability in both Denmark and UK because sensory and chemical analysis will be conducted in Denmark while expert tasting will be conducted in both Denmark and UK.

Participants will be trained first on basic taste samples and model wines, then on real wines (Peyvieux & Dijksterhuis, 2001). On the final evaluation day, each participant will evaluate the 6 wine samples in triplicate. Training is deemed to be sufficient if the curves for replicates are not different for more than 40% of the time (Peyvieux & Dijksterhuis, 2001)

- Metrics: Area under the curve (AUC), maximum intensity ($I_{\max}$), time to $I_{\max}$ ($T_{\max}$). See figure below for typical TI curve and common parameters (Goodstein et al., 2014).



WP4: Identifying common acidity descriptors via an online questionnaire

- Location: Online questionnaire
- Participants: MWs and MW students recruited from the candidate's personal network and the MW and MW student newsletter. Inclusion criteria: participants from at least three geographical areas (e.g., Europe, N America, Asia) and a balance between MWs graduating before vs. after 2020 (50-50 if possible, at least 20 per group). Target N=60

(power calculation with $\alpha = 0.05$, power = 0.8, and large effect Cohen's $W = 0.5$ with 5 degrees of freedom yields 52 participants).

- Methodology: An online questionnaire to gather typical acidity descriptors for dry Riesling and CB from the four representative regions identified in WP1. The most common 3-4 descriptors for each variety will be combined to produce a final list of 6-8 adjective options for WP5.
- Software platform: Compusense or Qualtrics

WP5: Expert tasting

- Location: UCPH Food sensory lab, WSET School London
- Participants: Experts (defined as holding DipWSET or higher qualifications) recruited from wine schools around Copenhagen (if needed: London). Target: N=50 (power calculation with small to medium effect size partial $\eta^2 = 0.04$ yields 42 participants).
- Wines: 6 samples from WP1 served in ISO glasses in 25 mL portions
- Procedure: Participants will taste the samples twice over two blocks, first wearing nose clips to remove the effect of retronasal olfaction, then without nose clips for full flavour evaluation. For each wine, participants will be given the same list of 6-8 descriptors (identified from WP4) and rate on an unmarked visual analogue scale how much each descriptor matches the wine's acidity (anchors: not at all, extremely well).
- Software platform: Compusense or Qualtrics.

WP6: Data analysis and writeup

- Software: R, SPSS
- RQ1

- Analysis 1: Using results from **WP4**, run a chi-square goodness of fit test to assess whether specific descriptors were spontaneously named at greater than chance levels for specific wines. Visualise results with Correspondence Analysis (CA).
- Analysis 2: Using results from **WP5**: ANOVA with adjective and grape variety as independent variables and adjective rating as dependent variable to test which descriptors have significantly higher ratings for each grape variety. Additionally, run principal component analysis (PCA) to visualise connections between wine samples and adjective ratings.

- RQ2

- Analysis 1: ANOVA with grape variety as independent variable and various TI metrics (see **WP3**) as dependent variable to assess overall differences between grape varieties in terms of temporal acidity perception.
- Analysis 2: Using data from **WP2** and **WP3**, elucidate the relationship between chemical composition and TI metrics using correlation analysis.

- RQ3

- Analysis: For each descriptor, use multivariate regression analysis to predict the descriptor-matching score (**WP5**) based on each wine's chemical analysis (**WP2**), sensory evaluation (**WP3**), and tasting condition (noseclip vs. full retronasal). PCAs will be run to visualise multivariate relationships between descriptors and chemical+sensory data.

Potential to Contribute to the Body of Knowledge on Wine: Explain how this Research Paper will add to the current body of knowledge on this subject. (No more than 150 words)

While acidity is a key structural component of wine, there is little scientific support for how people use descriptors to communicate acidity. While people may intuitively use specific acid descriptors, such as *broad* for CB and *jagged* for SB, it is unknown what lies behind such

crossmodal associations. This research will link language use with sensory perception and chemical composition, therefore enriching our understanding of wine communication. From a theoretical perspective, this research contributes to the study of wine cognition by uncovering implicit drivers behind descriptor choice.

From a practical perspective, any evidence on variety-specific *acid shapes* will be a great help for those learning blind tasting. Moreover, this research may lead to greater focus placed on the temporal assessment of wine structure in wine education, going beyond absolute static attributes and adding more nuance to the way we currently evaluate wines.

Proposed Time Schedule/Programme: This section should layout the time schedule for the research, analysis and write-up of the Research Paper and should indicate approximate dates with key deliverables. *Dates of submission to both Advisors and the IMW must be those specified by the IMW.*

	Jan	Feb	March	April	May	June	July	Aug	Sept	Oct	Nov
WP1											
WP2											
WP3											
WP4											
WP5											
WP6											

WP1: Wine selection and sourcing (Jan-Feb)

- Key deliverable: Wines sourced by February 15

WP2: Chemical analysis (March)

- Key deliverable: Chemical analysis conducted by March 15

WP3: Sensory assessment by trained panel (Feb-April)

- Key deliverable:
 - Panel recruited by March 15
 - Panel trained and data collected by April 30

WP4: Questionnaire on common acidity descriptors for each grape variety (March – April)

- Key deliverable:
 - Questionnaire deployed by March 15
 - Questionnaire completed by April 15

WP5: Expert evaluation (May – July)

- Key deliverable:
 - Data collection completed by July 31

WP6: Data analysis and writeup (July – Dec)

- Key deliverable:
 - Submit RP to RP adviser November 1
 - Submit RP to institute December 9